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24 June 2004

DEPARTMENT OF DEFENSE HANDBOOK

RESISTORS,

SELECTION AND USE OF

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FORWARD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. This handbook provides selected standard resistors for use in the design of Department of Defense equipment. This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply.

   a. The application information and performance characteristics contained in this handbook are offered for guidance and are not to be considered as mandatory. Additional application information will be added when coordinated with the Department of Defense.

   b. Additional resistor types of this handbook will be developed as standard resistors of a given specification family are selected and coordinated with the Department of Defense.

3. Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAT, Post Office Box 3990, Columbus, Ohio 43218-3990 or by email Resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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1. SCOPE

1.1 Scope. This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply. This handbook consists of the following:

a. Selected standard resistor types, for use in the design and manufacturer of Department of Defense equipment under the jurisdiction of the Department of Defense.

b. Guides for the choice and application of resistors for use in Department of Defense equipment.

Requirements for resistors listed in this handbook are covered in the applicable specification (see 2.1). When it has been determined that circuit requirements cannot be met by using resistor styles or characteristics listed in the applicable specifications, the design engineer should, with the approval of the cognizant activity, select from the applicable resistor specification styles or characteristics not listed herein.

1.2 Purpose of handbook.

a. To provide the equipment designer with a selection of standard resistors for use in most Department of Defense applications.

b. To control and minimize the variety of resistors used in Department of Defense equipment in order to facilitate logistic support of equipment in the field.

c. To outline criteria pertaining to the use, choice, and application of resistors in Department of Defense equipment.

1.3 DSCC SpecFinder. DSCC SpecFinder is an online tool that can be used in conjunction with this handbook. The DSCC SpecFinder is a search tool to help you find standardization documents by selecting device characteristics. If you require certain performance features but don't know which documents might meet your needs, the DSCC Specification Finder will help you narrow your search down to a few candidate documents. DSCC SpecFinder may be found at: http://www.dscc.dla.mil/Programs/SpecFind/ .

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4 and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19	-	Resistor, Variable, Wire Wound (Low operating Temperature), General Specification For.
MIL-PRF-22	-	Resistor, Variable, (Wire Wound, Power Type), General Specification For.
MIL-PRF-26	-	Resistors, Fixed, Wire Wound (Power Type), General Specification For.
MIL-PRF-29	-	Resistors, Fixed, Meter Multiplier, External (High Voltage, Ferrule Terminal Type) General Specification For.

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MIL-PRF-94	-	Resistor, Variable, Composition, General Specification For.
MIL-PRF-914	-	Resistor Networks, Fixed, Film, Surface Mount, Nonestablished Reliability, and Established Reliability, General Specification For.
MIL-PRF-12934	-	Resistor, Variable, Precision, General Specification For.
MIL-PRF-18546	-	Resistors, Fixed, Wire Wound (Power Type, Chassis Mounted), General Specification For.
MIL-PRF-22097	-	Resistor, Variable, Nonwire Wound (Adjustment Type), General Specification For.
MIL-PRF-22684	-	Resistors, Fixed, Film (Insulated), General Specification For.
MIL-PRF-23648	-	Resistor, Thermal (Thermistor) Insulated, General Specification For.
MIL-PRF-27208	-	Resistor, Variable, Wire Wound, Nonprecision, General Specification For.
MIL-PRF-32159	-	Resistors, Chip, Fixed, Film, Zero Ohm, Industrial, High Reliability, Space Level, General Specification For.
MIL-PRF-32192	-	Resistors, Chip, Thermal (Thermistor), General Specification for
MIL-PRF-39002	-	Resistors, Variable, Wire Wound, Semi-Precision, General Specification For.
MIL-PRF-39005	-	Resistors, Fixed, Wire Wound (Accurate), Nonestablished Reliability, Established Reliability, General Specification For.
MIL-PRF-39007	-	Resistors, Fixed, Wire Wound (Power Type), Nonestablished Reliability, Established Reliability, and Space Level, General Specification For.
MIL-PRF-39009	-	Resistors, Fixed, Wire Wound (Power Type, Chassis Mounted), Nonestablished Reliability, and Established Reliability, General Specification For.
MIL-PRF-39015	-	Resistors, Variable, Wire Wound, Nonestablished Reliability, and Established Reliability, General Specification For.
MIL-PRF-39017	-	Resistors, Fixed, Film (Insulated), Nonestablished Reliability, and Established Reliability, General Specification For.
MIL-PRF-39023	-	Resistors, Variable, Nonwire Wound, Precision, General Specification For.
MIL-PRF-39035	-	Resistor, Variable, Nonwire Wound (Adjustment Type), Nonestablished Reliability, and Established Reliability, General Specification For.
MIL-PRF-49462	-	Resistors, Fixed, Film, High Voltage, General Specification For.
MIL-PRF-49465	-	Resistors, Fixed, Metal Element (Power Type), (Very Low Resistance Values), General Specification For.
MIL-PRF-55182	-	Resistors, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level, General Specification For.

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MIL-PRF-55342	-	Resistors, Fixed, Film, Chip, Nonestablished Reliability, Established Reliability, Space Level, General Specification For.
MIL-PRF-83401	-	Resistor Networks, Fixed, Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Film Resistors, General Specification For.
MIL-PRF-83530	-	Resistors, Voltage Sensitive (Varistor, Metal-Oxide), General Specification For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Rating and design application terms. A list of common terms used in rating and design application of resistors is as follows:

a. Ambient operating temperature. The temperature of the air surrounding an object, neglecting small localized variations.

b. Contact resistance variation. The apparent resistance seen between the wiper and the resistance element when the wiper is energized with a specified current and moved over the adjustment travel in either direction at a constant speed. The output variations are measured over a specified frequency bandwidth, exclusive of the effects due to roll-on or roll-off of the terminations and is expressed in ohms or percent of total nominal resistance.

c. Critical value of resistance. For a given voltage rating and a given power rating, there is only one value of resistance that will dissipate full rated power at rated voltage. This value of resistance is commonly referred to as the “critical value of resistance.” For values of resistance below the critical value, the maximum (element) voltage is never reached and, for values of resistance above critical value, the power dissipated becomes lower than rated. Figure 1 shows this relationship.

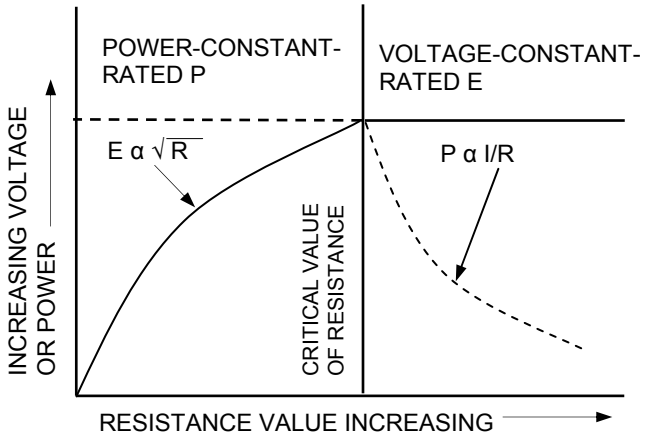


FIGURE 1. Maximum working voltage and critical value of resistance.

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d. Dielectric strength. The ultimate breakdown voltage of the dielectric or insulation of the resistor when the voltage is applied between the case and all terminals tied together. Dielectric strength is usually specified at sea level and simulated high altitude air pressures.

e. Hot-spot temperature. As defined in Department of Defense specifications, the maximum temperature measured on the resistor due to both internal heating and the ambient operating temperature. Maximum hot-spot temperature is predicated on thermal limits of the materials and the design. The hot-spot temperature is also usually established as the top temperature on the derating curve at which the resistor is derated to zero power.

f. Insulation resistance. The dc resistance measured between all terminals connected together and the case, exterior insulation, or external hardware.

g. Maximum (element) working voltage ($E = \sqrt{PR}$). The maximum voltage stress (dc or rms) that may be applied to the resistor (resistance element) is a function of (1) the materials used, (2) the required performance, and (3) the physical dimensions. (See figure 1)

h. Noise. An unwanted voltage fluctuation generated within the resistor. Total noise of a resistor always includes Johnson noise 1/ which is dependent only on the resistance value and temperature of the resistance element. Depending on the type of element and construction, total noise may also include noise caused by current flow, and noise caused by cracked bodies and loose end caps or leads. For variable resistors, noise may also be caused by jumping of contact over turns of wire and by an imperfect electrical path between the contact and resistance element.

i. Resistance temperature characteristic (temperature coefficient). The magnitude of change in resistance due to temperature, usually expressed in percent per degree Celsius or parts per million per degree Celsius (ppm/^{O}C). If the changes are linear over the operating temperature range, the parameter is known as “temperature coefficient”.

j. Resistance tolerance. The permissible deviation of the manufactured resistance value (expressed in percent) from the specified nominal resistance value at standard (or stated) environmental conditions.

k. Stability. The overall ability of a resistor to maintain its initial resistance value over extended periods of time when subjected to any combination of environmental conditions and electrical stresses.

4. GENERAL REQUIREMENTS

4.1 Choice of resistor types. The variety of resistor types used in any particular equipment should be the minimum necessary to obtain satisfactory performance. Where more than one type of resistor may be used in a given application (such as, fixed, film, insulated versus fixed, film, insulated (high stability)), consideration should be given to cost and availability (use of strategic materials, multiple sources). The resistors identified in this handbook meet all the criteria for standard types (see 1.1 and 4.4).

4.1.1 Reliability. Where quantitative reliability requirements specified as part of the equipment requirements are such that the use of parts with established reliability is dictated, such parts should be selected from the established reliability specification.

4.1.2 Qualified sources. After a preliminary selection of the desired resistor has been made, reference should be made to the applicable qualified products list for listing of qualified sources.

4.2 Item identification. A type designation for any resistor referenced herein may be constructed as indicated in the example given in the applicable section. The Part Identification Number (PIN) designations are depicted in the applicable specification.

1/ Johnson, J. B., “Thermal Agitation of Electricity in Conductors,” Physical Review, volume 32 (July, 1928, 97-109).

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4.3 Conflict of requirements. This handbook provides selected standard resistors for use in the design of Department of Defense equipment. This handbook is for guidance only. This handbook can not be sited as a requirement.

4.4 Criteria for inclusion in this handbook. The criteria for the inclusion of resistor types in this handbook are as follows:

a. The resistor should be the best type available for general use in military equipment.

b. Coordinated Department of Defense specifications should be available (see 2.1).

c. Resistors should be in production, or should have been in production.

5. DETAILED REQUIREMENTS

5.1 Detailed requirements. The detailed requirements for standard resistor types are contained in the applicable specification of this handbook.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. General application notes are as indicated in the appendix.

6.2 Subject term (key word) listing.

Adjustable	
Chip	Space level
Established reliability	Surface mount
Film	Thermistor
Fixed	Trimmer
Lead-screw	Variable
Network	Varistor
Nonestablished reliability	Wirewound
Nonwire wound	

6.3 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website http://www.epa.gov/osw/hazard/wastemin/priority.htm. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (See Section 3).

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

GENERAL APPLICATION INFORMATION

A.1. SCOPE

A.1.1 Scope. The application information in this handbook is designed to help in the selection of specified resistors (application information pertaining to specific resistors types is contained in the applicable sections). As with other types of components, the most important thing a user must decide is which of the numerous types of resistors will be best for use in the military equipment being designed. Proper selection in its broadest sense is the first step in building reliable equipment. To properly select the resistors to be used, the user must know as much as possible about the types from which to choose. The advantages and disadvantages should be known, as well as their behavior under various environmental conditions, their construction, and their effect on circuits and the effect of circuits on them, and a knowledge of what makes resistors fail. This appendix is not a mandatory part of the handbook. The information contained herein is intended for guidance only.

A.1.1.1 Resistor types. All variable and fixed resistors, of the types widely used in electronic equipment, can be grouped into one of three basic types. They are “composition” types, “film” types, or “wirewound” types. As the name indicates, the “composition” type is made of a mixture of resistive material and a binder which are molded into the proper shape and resistance value. The “film” type is composed of a resistive film deposited on, or inside of, an insulating cylinder or filament. The “wirewound” type is made up of resistance wire, wound on an insulated form. These basic types differ from each other in size, cost, resistance range, power rating, and general characteristics. Some are better than others for particular purposes; no one type has all of the best characteristics. The choice among them, therefore, depends on the requirements, both initial and long-term; the environment in which they must exist; and numerous other factors which the designer must understand. The designer must realize that the summaries of the requirements of a particular application must be taken into consideration and compared with the advantages and drawbacks of each of the several types, before a final choice is made. Tables A-I, A-II, A-III, A-IV, A-V and A-VI provide a selection guide for fixed and variable resistors included in this handbook.

The Department of Defense resistor specification categories are shown in figure A-1.

A.1.1.2 Resistance Tolerances. Resistance tolerances will be coded as follows:

Code	Tolerance	Code	Tolerance	Code	Tolerance
V	0.0005	B	0.1	H	3.0
T	0.01	D	0.5	J	5.0
Q	0.02	F	1.0	K	10.0
A	0.05	G	2.0		

A.2. APPLICABLE DOCUMENTS.

A.2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4 and 5 of this specification, whether or not they are listed.

A.2.2 Government documents.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-R-93	-	Resistors, Fixed, Wire-Wound (Accurate), General Specification for Inactive for New Design
MIL-R-10509	-	Resistors, Fixed, Film (High Stability), General Specification For Inactive for New Design

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APPENDIX A

MIL-R-39008 - Resistors, Fixed, Composition (Insulated) Established Reliability, General Specification For - CANCELLED

A.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-B545 - Standard Specification for Electrodeposited Coatings of Tin.

(Copies of this document are available online at http://www.astm.org/ or should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, Pennsylvania 19428-2959.)

A.3. GENERAL CHARACTERISTICS OF RESISTORS

A.3.1 General characteristics of fixed resistors.

A.3.1.1 Fixed, Film Resistors. See table A-I.

A.3.1.1.1 MIL-PRF-22684, RL42...TX, Resistors, Fixed Film (Insulated). These fixed, film, resistors have semi-precision characteristics and small sizes. The sizes and wattage rating are comparable to those of MIL-R-39008 (cancelled) and stability is between MIL-R-39008 and MIL-PRF-55182. Design parameter tolerances are looser than those of MIL-PRF-55182 but good stability makes them desirable in most electronic circuits. These resistors are capable of full load operation at an ambient temperature of +70°C and have a resistance-temperature characteristic of ±200 parts per million per degree Celsius (ppm/^{O}C). See MIL-PRF-39017.

NOTE: MIL-PRF-22684 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.2 MIL-PRF-32159, RCZ, Resistors, Chip, Fixed, Film, Zero Ohm, Industrial, High Reliability, Space Level. These fixed, film, chip resistors are primarily intended for incorporation into surface mount applications. This specification has three product levels, a space level “T” part number with 100 percent burn-in and screening, a high reliability 100 percent burn-in screened “M” part number level, and a part level “C” using the manufacturer’s inspection system to validate conformance.

A.3.1.1.3 MIL-PRF-39017, RLR, Resistors, Fixed Film (Insulated), Nonestablished Reliability and Established Reliability. These fixed, film, resistors have semi-precision characteristics and small sizes. The sizes and wattage rating are comparable to those of MIL-R-39008 (cancelled) and stability is between MIL-R-39008 and MIL-PRF-55182. Design parameter tolerances are looser than those of MIL-PRF-55182 but good stability makes them desirable in most electronic circuits. These resistors are capable of full load operation at an ambient temperature of +70°C and have a resistance-temperature characteristic of ±100 ppm/^{O}C and ±350 ppm/^{O}C. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests.
Replaces MIL-PRF-22684 inactive specification sheets.

NOTE: MIL-PRF-39017 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.4 MIL-PRF-49462, RHV, Resistors, Fixed, Film, High Voltage. These resistors are intended for use in electronic circuits where high voltages and high resistance values are used.

NOTE: MIL-PRF-49462 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

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A.3.1.1.5 MIL-PRF-49465, RLV, Resistor, Fixed, Metal Element (Power Type), (Very Low Resistance Value). These are power type, very low resistance values (1 ohm and below), fixed resistors (2 terminal and 4 terminal) for use in electrical, electronic, communications, and associated equipment. Included are precision resistors of 1, 3, and 5 percent, and 5 and 10 percent initial resistance tolerances with power ratings ranging from 2 watts to 10 watts at +25°C derated to 0 watts at +275°C

NOTE: MIL-PRF-49465 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.6 MIL-PRF-55182, RNC, RNN, or RNR, Resistor, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level. These fixed, film resistors including both hermetically and nonhermetically sealed types possess a high degree of stability, with respect to time, under severe environmental conditions, with an established reliability. Use in circuits requiring higher stability than provided by composition resistors or film, insulated, resistors, and where ac frequency requirements are critical. Operation is satisfactory from dc to 100 megahertz (MHz). Metal films are characterized by low temperature coefficients and are usable for ambient temperatures of +125°C or higher with small degradation. The resistors have product levels ranging from Non-ER (C), a life failure rate (FR) 1.0 (M) percent to 0.001 (S) percent per 1,000 hours and space level (T). The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-R-10509 RN, fixed, film (high stability) inactive specification sheets.

NOTE: MIL-PRF-55182 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.7 MIL-PRF-55342, RM, Resistor, Chip, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level. These chip resistors are primarily intended for incorporation into hybrid microelectronic circuits. They are designed for use in critical circuitry where stability, long life, reliable operation, and accuracy are of prime importance. These resistors are uncased, leadless chip devices and possess a high degree of stability with respect to time, under severe environmental conditions. These resistors provide life failure rates ranging from 1.0 percent to 0.001 percent per 1,000 hours. The failure rates are established at 60 percent confidence level on the basis of life tests. The resistors have product levels ranging from Non-ER (C), a life failure rate (FR) 1.0 (M) percent to 0.001 (S) percent per 1,000 hours and space level (T). The FR levels are established at a 60 percent confidence level on basis of life tests.

NOTE: MIL-PRF-55342 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

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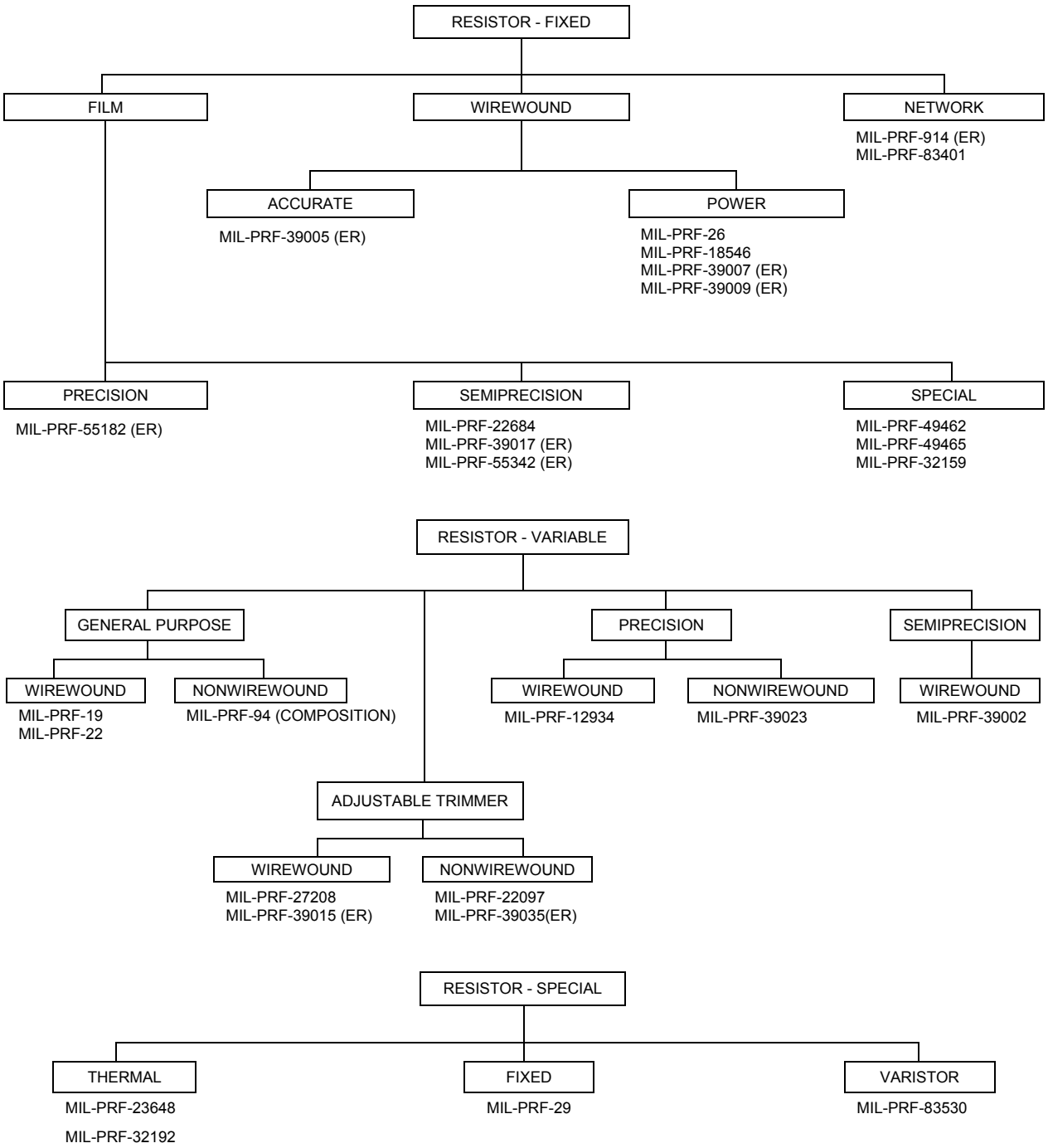


FIGURE A-I. Department of Defense resistor specification categories.

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TABLE A-I. Fixed film resistor selection guidance table.

Specification	Styles available	Power and max voltage ratings	Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and Configuration (see figure A-2)
MIL-PRF-22684 Insulated	RL42…TX	2 W 500 V	G and J	10 to 1.5 M	±200	0.728 x 0.336 A

Specification	Styles available	Power ratings	Termination	Current rating (in amps)	Max resistance value (in ohms)	Maximum body size (inches) and Configuration (see figure A.2)
MIL-PRF-32159 Chip, zero ohm, industrial, high reliability, space level	RCZ0502	0.050	B,G,W	1.30	0.030	0.061 X 0.030 T
			C,D	0.91	0.060	
			U,T	0.58	0.150	
	RCZ0505	0.100	B,G,W	2.20	0.020	0.061 X 0.055 T
			C,D	1.80	0.030	
			U,T	1.00	0.100	
	RCZ1005	0.200	B,G,W	2.80	0.025	0.112 X 0.055 T
			C,D	2.20	0.040	
			U,T	1.30	0.125	
	RCZ1505	0.150	B,G,W	2.10	0.035	0.162 X 0.055 T
			C,D	1.70	0.050	
			U,T	0.93	0.175	
	RCZ2208	0.225	B,G,W	2.50	0.035	0.237 X 0.080 T
			C,D	2.10	0.050	
			U,T	1.10	0.175	
	RCZ0705	0.150	B,G,W	2.70	0.020	0.086 X 0.055 T
			C,D	2.20	0.030	
			U,T	1.20	0.100	
	RCZ1206	0.225	B,G,W	3.20	0.025	0.134 X 0.068 T
			C,D	2.50	0.040	
			U,T	1.40	0.125	
	RCZ2010	0.800	B,G,W	5.70	0.025	0.218 X 0.103 T
			C,D	4.50	0.040	
			U,T	2.50	0.125	
	RCZ2512	1.000	B,G,W	6.30	0.025	0.268 X 0.129 T
			C,D	5.00	0.040	
			U,T	2.80	0.125	
	RCZ1010	0.500	B,G,W	5.00	0.020	0.112 X 0.105 T
			C,D	4.10	0.030	
			U,T	2.20	0.100	
	RCZ0402	0.040	B,G,W	1.20	0.030	0.050 X 0.027 T
			C,D	0.82	0.060	
			U,T	0.52	0.150	
	RCZ0603	0.070	B,G,W	1.50	0.030	0.070 X 0.037 T
			C,D	1.10	0.060	
			U,T	0.68	0.150	
	RCZ0302	0.035	B,G,W	1.10	0.030	0.038 X 0.027 T
			C,D	0.76	0.060	
			U,T	0.48	0.150	

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TABLE A-I. Fixed film resistor selection guidance table - Continued.

Specification	Styles available	Power and max voltage ratings	Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and Configuration (see figure A-2)
MIL-PRF-39017 Insulated, nonestablished and established reliability	RLR05	0.125 W 200 V	F, G, J and K	2.7 to 1 M 1.1 M to 22 M	±100 ±350	0.170 x 0.074 A
	RLR07	0.25 W 250 V		1 to 10 M 11 M to 22 M		0.281 x 0.098 A
	RLR20	0.5 W 350 V		1 to 3.01 M 3.3 M to 22 M		0.416 x 0.161 A
	RLR32	1 W 500 V		1 to 2.7 M 3.0 M to 22 M		0.593 x 0.205 A
MIL-PRF-49462 High voltage	RHV30 RHV31 RHV32 RHV33 RHV34 RHV35	0.25 W/ 750 V 0.5 W/ 1.5 kV 1.0 W/ 3.0 kV 2.0 W/ 5.0 kV 3.0 W/ 10.0 kV 5.0 W/ 20.0 kV	F, G and J	100 k to 100 M 100 k to 392 M 49.9 k to 499 M 100 k to 499 M 200 k to 1 G 330 k to 1 G	R < 500 MΩ RTC ≤ 200 ppm R ≥ 500 MΩ RTC ≤ 500 ppm	0.306 x 0.098 0.431 x 0.154 0.752 x 0.328 1.124 x 0.328 2.124 x 0.328 3.124 x 0.328 A

Specification	Styles available	Power ratings	Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and Configuration (see figure A-2)
MIL-PRF-49465 Metal element, power type, very low resistance values	RLV10	5 W	F, H and J	0.01 to 0.0249 0.025 to 0.0499 0.05 to 0.0749 0.075 to 0.099 0.1 to 0.5	±150 ±125 ±100 ±50 ±50	0.999 x 0.406 O
	RLV30	3 W		0.01 to 0.0249 0.025 to 0.0499 0.05 to 0.0749 0.075 to 0.099 0.1 to 0.2	±350 ±200 ±125 ±75 ±50	0.591 x 0.236 A
	RLV31	5 W		0.01 to 0.0249 0.025 to 0.0499 0.05 to 0.0749 0.075 to 0.099 0.1 to 0.3	±250 ±150 ±100 ±75 ±50	.956 x .361 A

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TABLE A-I. Fixed film resistor selection guidance table - Continued.

Specification	Styles available	Power and max voltage ratings	Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and Configuration (see figure A-2)
MIL-PRF-55182 Nonestablished and established reliability, space level	RNR/N/C50	0.05 W 200 V	B, D and F	10 to .796 M	±25, ±50, ±100	0.180 x 0.080 A
	RNR/N/C55	0.1 W 200 V		10 to 2.0 M		0.281 x 0.140 A
	RNR/N/C60	0.125 W 250 V		1.0 to 4.02 M		0.437 x 0.165 A
	RNR/N/C65	0.25 W 300 V		1.0 to 8.06 M		0.656 x 0.250 A
	RNR/N/C70	0.5 W 350 V		1.0 to 15 M		0.875 x 0.328 A
	RNR/C75	1 W 750 V		10 to 20 M	±25	1.24 x 0.437 A
	RNC90	0.3 W 300 V	V, T, A, B, D and F	4.99 to 200k	±5 ±10	0.320 x 0.336 x 0.120 K
MIL-PRF-55342 Chip nonestablished and established reliability, space level	RM0302	0.040 W 15 V	B, F, G, J and K	1 to 22M	±25, ±50 ±100, ±300 ±25, ±50 ±100, ±300	0.032 x 0.022 x 0.010/0.025 T
	RM0402	0.050 W 25 V				0.041 x 0.022 x 0.010/0.033 T
	RM0502	0.05 W 40 V				0.055 x 0.035 x 0.010/0.030 T
	RM0505	0.125W 40 V				0.05 x 0.05 x 0.012/0.033 T
	RM0603	0.1 W 50 V				0.060 x 0.032 x 0.010/0.033 T
	RM0705	0.150 W 50V				0.075 x 0.05 x 0.015/0.033 T
	RM1005	0.2 W 50 V				0.10 x 0.05 x 0.015/0.033 T
	RM1010	0.5 W 75 V				0.100 x 0.100 0.015/0.033 T
	RM1206	0.25 W 100V				0.126 x 0.063 x 0.015/0.033 T
	RM1505	0.150 W 40 V				0.15 x 0.05 x 0.015/0.033 T
	RM2010	0.8 W 150 V				.206 x .098 x .015/.033 T
	RM2208	0.225 W 40 V				.230 x .085 x .015/.033 T
	RM2512	1 W 200 V				248 x .124 x .015/.033 T

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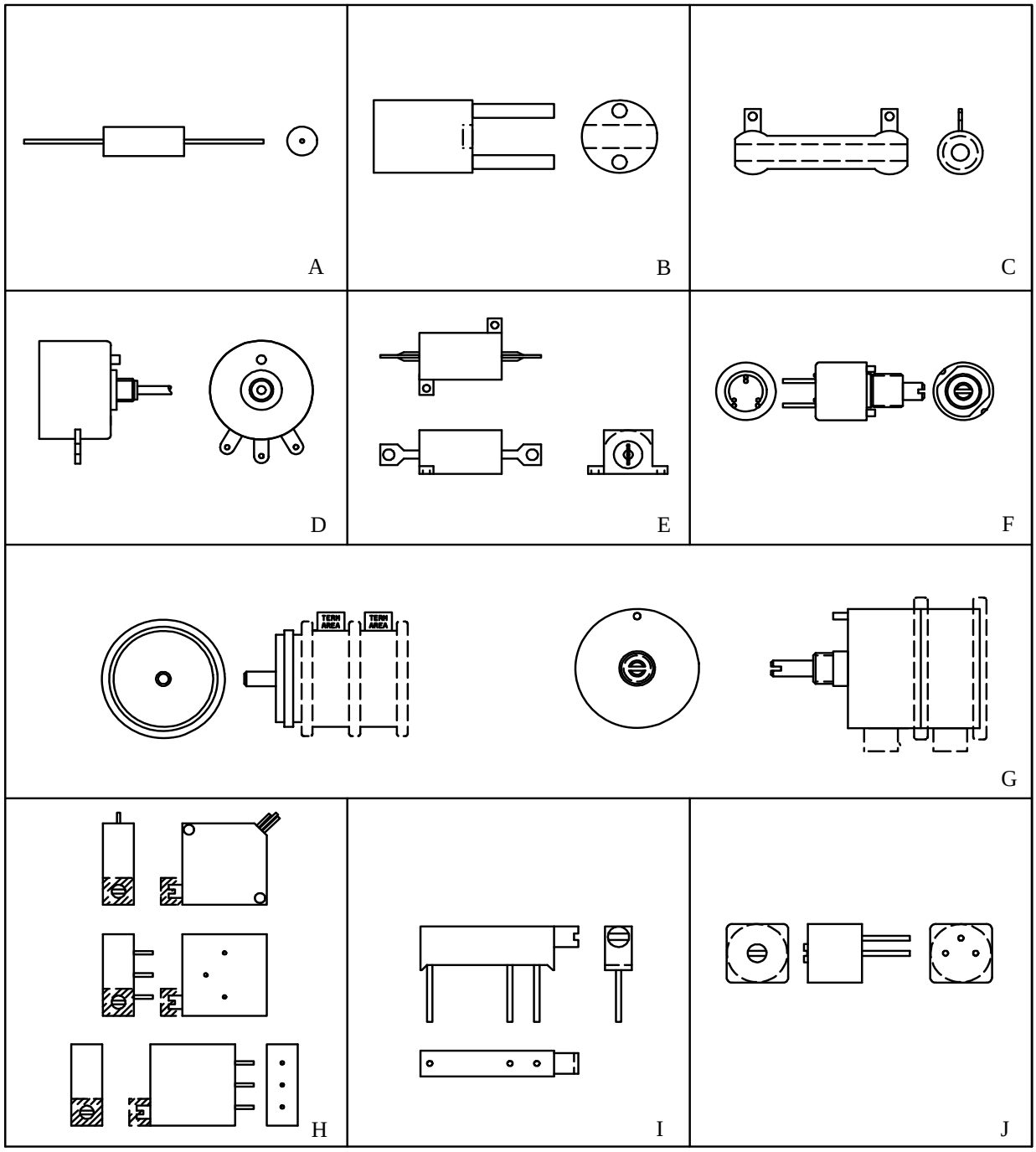


FIGURE A-2. Configurations.

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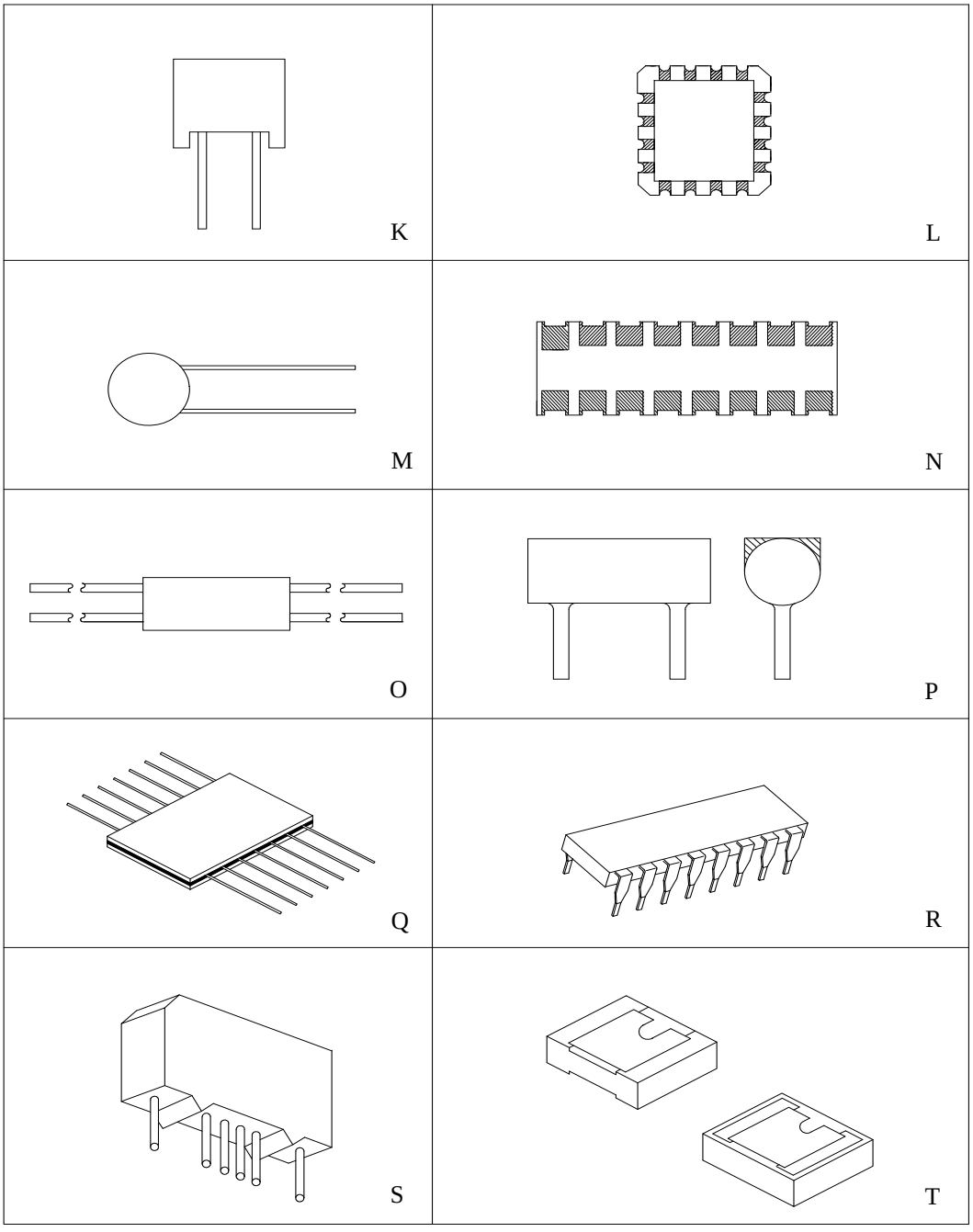


FIGURE A-2. Configurations. - Continued.

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A.3.1.2 Fixed Wire-wound Resistors. See table A-II.

A.3.1.2.1 MIL-PRF-26, RW, Resistors, Fixed, Wire Wound (Power Type). These power type, wire wound fixed resistors are for use in electrical, communication and associated equipment. Included are general purpose styles of 5 percent initial resistance tolerance with power ratings ranging from 3 watts to 240 watts at +25°C, derated to 0 power at 350°C and precision axial lead types of 0.1 percent, 0.5 percent, and 1 percent initial resistance tolerance with power ranging from 1 watt to 10 watts at +25°C, derated to 0 power at +275°C. Use where large power dissipation is required and where ac performance is relatively unimportant (such as, when used as voltage divider or bleeder resistors in dc power supplies, or for series dropping). They are generally satisfactory for use at frequencies up to 20 kilohertz (kHz) even though the ac characteristics are controlled. Neither the wattage rating nor the rated continuous working voltage may be exceeded. See MIL-PRF-39007.

A.3.1.2.2 MIL-PRF-18546, RE, Resistors, Fixed, Wire Wound (Power Type, Chassis Mounted). These fixed, wire wound resistors are established reliability chassis mounted, power type units. They utilize the principle of heat dissipation through a metal mounting surface. These resistors are capable of full load operation at an ambient temperature of +25°C when mounted on a specified chassis area. Use where power tolerance and relatively large power dissipation is required for a given unit size than is provided by MIL-PRF-26 resistors, where ac performance is noncritical (such as, voltage divider or bleeder resistors in dc power supplies or series-dropping circuits). See MIL-PRF-39009.

A.3.1.2.3 MIL-PRF-39005, RBR, Resistors, Fixed, Wire Wound, (Accurate), Nonestablished Reliability, Established Reliability. These nonestablished reliability, established reliability, accurate, wire wound fixed resistors that have a maximum resistance tolerance of 1 percent and a high degree of stability with respect to time under specified environmental conditions. These resistors are suitable for continuous full load (or voltage) operation at any ambient temperature up to +125°C, and when properly derated to +145°C. Use in circuits requiring higher stability than provided by composition or film resistors, and where ac frequency performance is not critical. Operation is satisfactory from dc to 50 kHz. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-R-93 inactive specification sheets.

A.3.1.2.4 MIL-PRF-39007, RWR, Resistors Fixed, Wire Wound (Power Type), Nonestablished Reliability, Established Reliability, and Space Level. These nonestablished reliability, established reliability, or space level, axial leaded precision power type, wire wound fixed resistors having a +25°C ambient operating temperature derated to 0 load at +250°C. Use where large power dissipation is required and where ac performance is relatively unimportant (such as, when used as voltage divider or bleeder resistors in dc power supplies, or for series dropping). They are generally satisfactory for use at frequencies up to 20 kilohertz (kHz) even though the ac characteristics are controlled. Neither the wattage rating nor the rated continuous working voltage may be exceeded. The resistors have product levels ranging from Non-ER (C), a life failure rate (FR) 1.0 (M) percent to 0.001 (S) percent per 1,000 hours and space level (T) The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-PRF-26 inactive specification sheets.

A.3.1.2.5 MIL-PRF-39009, RER, Resistors, Fixed, Wire Wound (Power Type, Chassis Mounted), Nonestablished Reliability, Established Reliability. These fixed, wire wound resistors are established reliability chassis mounted, power type units. They utilize the principle of heat dissipation through a metal mounting surface with full rated wattage at +25°C. These resistors should not be used in circuits where their ac performance is of critical importance, although provision have been made, in particular styles, to minimize inductance. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-PRF-18546 inactive specification sheets.

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TABLE A-II. Fixed wire wound resistor selection guidance table.

Specification	Styles available	Power ratings (in Watts)	Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and configuration (see figure A-2)
MIL-PRF-26 Power type	RW29 RW31 RW33 RW35 RW37 RW38 RW47	11 14 26 55 113 159 210	J and K	0.1 to 5.6 k 0.1 to 6.8 k 0.1 to 18 k 0.1 to 43 k 0.1 to 91 k 0.1 to 0.15 M 0.1 to 0.18 M	±650 (.1 ≤ R < .499) ±500 (.499 ≤ R < 1) ±400 (1 ≤ R < 20) ±10 (R ≥ 20)	1.812 x 0.500 1.562 x 0.594 3.062 x 0.594 4.062 x 0.906 6.062 x 1.312 8.062 x 1.312 10.562 x 1.312 C
	RW56	14	J and K	0.1 to 9.1 k	±400 (R < 20Ω) ±260 (R ≥ 20Ω)	2.094 x .563 A
MIL-PRF-18546 Power type, chassis mounted	RE77	75 120	F	0.05 to 29.4 k 0.1 to 35.7 k	±30 (R ≥ 2k) ±50 (R < 2k)	3.594 x 1.781 x 2.843 E
	RE80					4.594 x 2.219 x 3.031 E

Specification	Styles available	Power and max voltage ratings	Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and configuration (see figure A-2)
MIL-PRF-39005 Accurate, nonestablished and established reliability	RBR52	0.5 W 600 V	T, Q, A, B and F	0.1 to 1.21 M	±90 (R < 1) ±30 (1 ≤ R < 10) ±15 (10 ≤ R < 100) ±10 (R ≥ 100)	1.020 x 0.390 A
	RBR53	0.33 W 300 V		0.1 to 1.1 M		0.770 x 0.390 A
	RBR54	0.25 W 300 V		0.1 to .562 M		0.770 x 0.265 A
	RBR55	0.15 W 200 V		0.1 to .332 M		0.520 x 0.265 A
	RBR56	0.125 W 150 V		0.1 to .220 M		0.364 x 0.265 A
	RBR71	0.125 W 150 V		0.1 to .15 M		0.343 x 0.281 B
	RBR80	0.1 W 100 V	T, Q, A, B and F	10 to .120 M 10 to .250 M	±10 (R < 100) ±5 (R ≥ 100)	0.325 x 0.160 P
	RBR81					

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TABLE A-II. Fixed wire wound resistor selection guidance table - Continued.

Specification	Styles available	Power rating (in Watts)	Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and configuration (see figure A-2)
MIL-PRF-39007 Power type, nonestablished and established reliability, space level	RWR78	10	B, D and F	0.1 to 39.2 k	±650 (.1 ≤ R ≤ .499) ±400 (.499 < R ≤ 1) ±50 (1 < R ≤ 10) ±20 (R >10)	1.842 x 0.406 A
	RWR80	2		0.1 to 3.16 k		0.437 x 0.125 A
	RWR81	1		0.1 to 1 k		0.281 x 0.105 A
	RWR82	1.5		0.1 to 1.3 k		0.328 x 0.101 A
	RWR84	7		0.1 to 12.4 k		0.937 x 0.343 A
	RWR89	3		0.1 to 4.12 k		0.622 x 0.218 A
MIL-PRF-39009 Power type, chassis mounted, nonestablished and established reliability	RER40 RER45 RER50 RER55	5 10 20 30	F	1 to 1.65 k 1 to 2.80 k 1 to 6.04 k 1 to 19.6 k	±100 (R < 1) ±50 (1 ≤ R < 19.6) ±30 (R ≥ 20)	0.662 x 0.677 x 0.351 0.812 x 0.843 x 0.437 1.124 x 1.125 x 0.593 2.000 x 1.187 x 0.656 E
	RER60 RER65 RER70 RER75	5 10 20 30		0.1 to 3.32 k 0.1 to 5.62 k 0.1 to 12.1 k 0.1 to 39.2 k		0.662 x 0 .677 x 0.351 0.812 x 0.843 x 0.437 1.124 x 1.125 x 0.593 2.000 x 1.187 x 0.656 E

A.3.1.3 Fixed Film Networks. See table A-III.

A.3.1.3.1 MIL-PRF-914, RNS, Resistor, Network, Fixed, Film, Surface Mount, Nonestablished Reliability, and Established Reliability. These networks are primarily intended for use in surface mount applications where space is a major concern. These resistors can either be hermetically or nonhermetically sealed and consist entirely of fixed, film resistors. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests.

NOTE: MIL-PRF-914 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.3.2 MIL-PRF-83401, RZ, Resistor Networks, Fixed Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Film Resistors. These networks are designed for use in critical circuitry where stability, long life, reliable operation, and accuracy are of prime importance. They are particularly desirable for use where miniaturization is important and ease of assembly is desired. They are useful where a number of resistors of the same resistance value are required in the circuit. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests.

NOTE: MIL-PRF-83401 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

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TABLE A-III. Fixed film network resistor selection guidance table.

Specification	Styles available	Schematic	Power ratings		Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and configuration (see figure A-2)
			Characteristics					
			K and M	C, R, H, and V				
MIL-PRF-914 Surface mount, nonestablished and established reliability	RNS030	W M E S	0.10/0.80 0.05/0.75 0.10/0.80 0.05/0.70	0.050/0.40 0.025/0.40 0.050/0.40 0.025/0.35	B, D, F, G and J	10 to 2.2 M	±25, ±50, ±100, ±300	.300 x .300 x .050 .300 x .300 x .085 .300 x .300 x .090 L
	RNS040	P M E S	0.10/1.0 0.05/0.95 0.10/1.0 0.05/1.0	0.050/0.50 0.025/0.475 0.050/0.50 0.025/0.50				.350 x .350 x .050 .350 x .035 x .085 .350 x .350 x .090 L
	RNS050	A B C J	0.10/0.80 0.055/0.80 0.050/0.80 0.030/0.80	0.050/0.40 0.025/0.375 0.100/0.40 0.015/0.40				.150 x .410 x.035 N

Specification	Styles available	Schematic	Power ratings		Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and configuration (see figure A-2)
			Characteristics					
			H, K and M	C and V				
MIL-PRF-83401 Resistor Networks,.	RZ010	A B J	0.2/1.4 0.1/1.3 0.050/1.2	0.1/0.7 0.05/0.65 0.025/0.6	B, D, F, G and J	10 to 1 M	± 50, ±100, ±300	.785 x .305 x .200 R
	RZ020	A B J	0.2/1.6 0.1/1.5 0.050/1.4	0.1/0.8 0.05/0.75 0.025/0.7				.876 x .305 x .200 R
	RZ030	A B J	0.05/0.35	0.025/0.325 0.015/0.35				.385 x .305 x .075 Q
	RZ040	C G H	0.2/1.0 0.2/0.6 0.11/0.88	0.1/0.5 0.1/0.3 0.06/0.48	F, G and J			.598 x .098 x .350 S
	RZ050	C G H	0.2/1.4 0.2/0.8 0.11/1.32	0.1/0.7 0.1/0.4 0.06/0.72				.798 x .098 x .350 S
	RZ060	C G H	0.2/1.8 0.2/1.0 0.11/1.8	0.1/0.9 0.1/0.5 0.06/0.9				.998 x .098 x .352 S
	RZ070	C G H	0.12/0.6 0.12/0.36 0.07/0.6	0.06/0.3 0.06/0.18 0.04/0.3	B, D, F, G and J			.598 x .098 x .197 S
	RZ080	C G H	0.12/0.84 0.12/0.48 0.07/0.84	0.06/0.42 0.06/0.24 0.04/0.42				.798 x .098 x .197 S
	RZ090	C G H	0.12/1.08 0.12/0.60 0.07/1.08	0.06/0.54 0.06/0.30 0.04/0.54				798 x .098 x .197 S

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APPENDIX A

TABLE A-III. Fixed film network resistor selection guidance table - Continued.

Specification	Styles available	Schematic	Power ratings		Resistance tolerance (± percent) (see A.1.1.2)	Resistance range (ohms)	Resistance temperature coefficient (ppm/°C)	Maximum body size (inches) and configuration (see figure A-2)
			Characteristics					
			H, K and M	C and V				
MIL-PRF-83401 Resistor Networks	RZ100	A B J	0.05/0.4 0.025/0.375 0.015/0.42		B, D, F, G and J	10 to 1 M	±50, ±100, ±300	.410 x .305 x .075 Q
	RZ130	A B J	0.2/1.4 0.1/1.3 0.050/1.2	0.1/0.7 0.05/0.65 0.025/0.6				.785 x .305 x .200 R
	RZ140	A B J	0.2/1.6 0.1/1.5 0.050/1.4	0.1/0.8 0.05/0.75 0.025/0.7				.876 x .305 x .200 R
	RZ150	A B J	0.05/0.35 0.025/0.325 0.015/0.35					.385 x .305 x .075 Q
	RZ180	A001 A002	0.10 0.10	0.40 0.50	G	See Schematics		.598 x .098 x .350 S
	RZ190	A001 to A006, A011, A012 A007 to A009 A010	0.10/0.60 0.10/0.50 0.10/0.40					.798 x .098 x .197 S
	RZ210	C G H	0.12/0.6 0.12/0.36 0.07/0.6	0.06/0.3 0.06/0.18 0.04/0.3	B, D, F, G and J			.598 x .098 x .197 S
	RZ220	C G H	0.12/0.84 0.12/0.48 0.07/0.84	0.06/0.42 0.06/0.24 0.04/0.42				.798 x .098 x .197 S
	RZ230	C G H	0.12/1.08 0.12/0.60 0.07/1.08	0.06/0.54 0.06/0.30 0.04/0.54				.998 x .098 x .197 S
	RZ240	C G H	0.12/1.08 0.12/0.60 0.07/1.08	0.06/0.54 0.06/0.30 0.04/0.54				.998 x .098 x .197 S

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A.3.1.4 Variable Wire Wound Resistors. (See table A-IV)

A.3.1.4.1 MIL-PRF-19, RA, Resistor, Variable, Wire Wound (Low Operating Temperature). These variable resistors having a resistance element of wire wound on an insulating strip shaped in an arc, so that a contact bears uniformly on the resistance element when adjusted by a control shaft. These resistors are capable of full load operation at an ambient temperature of +40°C and are suitable for continuous operation when properly derated, at a maximum temperature of +105°C. Use primarily for noncritical, low power, low frequency applications where characteristics of wire wound resistors are more desirable than those of composition resistors.

A.3.1.4.2 MIL-PRF-22, RP, Resistor, Variable, Wire Wound (Power Type). These resistors have a resistance element of wire, wound linearly on an insulted strip shaped in an arc, such that a contact bears uniformly on the resistance element when adjusted by a control shaft. The power ratings cover a range from 6.25 watts to 1,000 watts, inclusive. Use in such applications as motor speed control, generator field control, lamp dimming, heater and oven control, potentiometer uses, and applications where variations of voltage and current are expected.

A.3.1.4.3 MIL-PRF-12934, RR, Resistor, Variable, Wire Wound, Precision. These precision, wire wound, variable resistors whose electrical output (in terms of percent of applied voltage) are linear or nonlinear with respects to the angular position of the shaft. These resistors are capable of full-load operation at maximum ambient temperatures of +70°C and +85°C and are suitable for continuous operation, when properly derated, to maximum temperatures of +125°C and +150°C. Use in servo mounting applications requiring precise electrical and mechanical output and performance. Used in computer, antenna, flight control, and bomb navigation systems.

A.3.1.4.4 MIL-PRF-27208, RT, Resistor, Variable, Wire Wound, Nonprecision. These wire wound variable resistors with a contact bearing uniformly over the entire surface of the entire resistive element, wound linearly, when positioned by an multiturn lead screw actuator. These resistors are capable of full-load operation at maximum ambient temperatures of +85°C and are suitable for continuous operation, when properly derated, to maximum temperatures of +150°C. Use for matching, balancing, and adjusting circuit variables in computers, telemetering equipment, and other critical applications. See MIL-PRF-39015.

A.3.1.4.5 MIL-PRF-39002, RK, Resistor, Variable, Wire Wound, Semi-Precision. These semi-precision, wire wound, variable resistors have a resistance element of wire, wound linearly on an insulated form shaped in an arc, so that a contact bears uniformly on the resistance element when adjusted by a contact shaft. The electrical output (in terms of percent of applied voltage) is linear with respect to angular position of the contact arm. These resistors are capable of full-load operation at maximum ambient temperatures of +85°C and are suitable for continuous operation, when properly derated, to maximum temperatures of +135°C. Use for matching, balancing, and adjusting circuit variables in computers, telemetering equipment, and other critical applications.

A.3.1.4.6 MIL-PRF-39015, RTR, Resistor, Variable, Wire Wound (Lead Screw Actuated), Nonestablished Reliability and Established Reliability. These nonestablished and established reliability lead screw actuated, wire wound variable resistors with a contact bearing uniformly over the surface of the entire resistive element, wound linearly, when position by a multiturn lead screw actuator. These resistors are capable of full load operation (when the maximum resistance is engaged), at maximum ambient temperature of +85°C and are suitable for continuous operation, when properly derated, at a maximum ambient temperature of +150°C. Use for matching, balancing, and adjusting circuit variables in computers, telemetering equipment, and other critical applications. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-PRF-27208 inactive specification sheets.

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APPENDIX A

TABLE A-IV. Variable wire wound resistor selection guidance table.

Specification	Styles	Power rating (Watts)	Resistance range (ohms)	Maximum body size (inches) Configuration (see figure A-2)
MIL-PRF-19 Low operating temperature	RA20 RA30	2, 1 4, 2.2	3 to 15k 3 to 25k	1.310 x .700 1.710 x .810 D
MIL-PRF-22 Power type	RP05 RP06 RP07 RP10 RP11 RP15 RP16 RP20 RP25 RP30 RP35 RP40 RP45 RP50 RP55	5 12.5 6.25 25 12.5 50 25 75 100 150 225 300 500 750 1,000	10 to 5k 1 to 3.5k 1 to 3.5k 1 to 5k 2 to 5k 1 to 10k 1 to 10k 2 to 10k 2 to 10k 2 to 10k 2 to 2.5k 2 to 2.5k 2 to 2.5k 2 to 2.5k 2 to 2.5k	0.525 x 0.687 0.906 x 0.751 1.094 x 1.126 1.680 x 1.410 1.880 x 1.750 2.410 x 1.440 2.750 x 1.750 2.810 x 1.780 3.190 x 1.780 4.060 x 2.030 5.090 x 2.160 6.090 x 2.410 8.090 x 2.250 10.090 x 3.030 12.310 x 3.250 D
MIL-PRF-12934 Precision	RR0900 RR1000 RR1100 RR1300 RR1400 RR2000 RR2100 RR3000 RR3100 RR3200 RR3300 RR3400 RR3500 RR3600 RR3700 RR3800 RR3900 RR4000 RR4100 RR2002 RR1004 2RR2104 RR3601 2RR3100	1.25 2.0 1.5 2.0 3.0 4.0 5.0 6.0 1.25 1.5 2.0 4.0 6.0 1.5 1.5 1.5 1.5 2.0 5.0 4.0 2.0 5.0 1.5 0.25mW	100 to 10K 100 to 50K 100 to 20K 100 to 40K 200 to 200K 100 to 60K 200 to 250K 200 to 100K 100 to 10K 100 to 20K 100 to 40K 100 to 60K 200 to 100K 100 to 50K 100 to 50K 100 to 100K 100 to 100K 100 to 50K 200 to 250K 20K 100K 20K / 20K 10K 665 / 665	0.880 x 0.812 0.880 x 1.625 1.067 x 0.812 1.468 x 1.062 1.468 x 2.250 2.031 x 1.312 2.031 x 2.250 3.031 x 1.312 0.906 x 0.750 1.093 x 0.750 1.468 x 1.062 2.031 x 1.156 3.031 x 1.156 0.906 x 1.076 0.906 x 1.076 0.906 x 1.219 0.906 x 1.219 0.890 x 1.500 1.844 x 2.094 2.005 x 1.781 0.880 x 1.580 2.005 x 4.311 0.922 x 0.875 1.062 x 0.765 G

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TABLE A-IV. Variable wire wound resistor selection guidance table. - Continued.

Specification	Styles	Power rating (Watts)	Resistance range (ohms)	Maximum body size (inches) (configuration) (see figure A-2)
MIL-PRF-27208 Nonprecision	RT12	0.75	10 to 20K	1.250 x 0.315 x 0.190 I
	RT24		10 to 10K	0.375 x 0.375 x0.195 H
	RT26	0.25	10 to 5K	0.250 x 0.250 x 0.165 H
MIL-PRF-39002 Semi-precision	RK09 RK11	1.21.5	10 to 100k 5k	0.515 x 0.650 0.615 x 0.315 F
MIL-PRF-39015 Lead screw actuated	RTR12	0.75	10 to 20K	1.250 x 0.315 x 0.190 I
	RTR22			0.500 x 0.500 x 0.235 H
	RTR24		10 to 10K	0.375 x 0.375 x0.195 H

A.3.1.5 Variable Nonwire Wound Resistors. See table A-V.

A.3.1.5.1 MIL-PRF-94, RV, Resistor, Variable Composition. These variable resistors have a composition resistance element shaped in an arc, and a contact bearing uniformly thereon, so that a change in resistance is produced between the terminal of the contact and the terminal of either end of the resistance element when the operating shaft is turned. These resistors are capable of full load operation (where maximum resistance is engaged) at a maximum ambient temperature of +70°C, and suitable for continuous operation when properly derated, at a maximum temperature of +120°C. Use where initial setting stability is not critical and long-term stability needs to be no better than ±20 percent.

A.3.1.5.2 MIL-PRF-22097, RJ, Resistor, Variable, Nonwire Wound (Adjustment Type). These multiturn lead screw actuated and single-turn nonwire wound variable resistors with a contact bearing uniformly over the entire surface of the entire resistive element, when positioned by the actuator. These resistors are capable of full load operation (where maximum resistance is engaged) at a maximum ambient temperature of +70°C and +85°C, and are suitable for continuous operation when properly derated, at a maximum temperature of +120°C and +150°C, respectively. Use for matching, balancing, and adjusting circuit variables in computers, telemetering equipment, and other critical applications. See MIL-PRF-39035.

A.3.1.5.3 MIL-PRF-39023, RQ, Resistor, Variable, Nonwire Wound Precision. These precision, nonwire wound, variable resistors whose electrical output (in terms of percent of applied voltage) are linear or nonlinear with respect to the angular position of the operating shaft. These resistors have a resistance tolerance of ±10 percent. These resistors are capable of full-load operation at maximum ambient temperatures of +70°C and are suitable for continuous operation, when properly derated, to maximum temperatures of +125°C. Use in servo mounting application requiring precise electrical and mechanical output and performance. Used in computer, antenna, flight control, and bomb navigation systems.

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APPENDIX A

A.3.1.5.4 MIL-PRF-39035, RJR, Resistors, Variable, Nonwire Wound (Adjustment Type), Nonestablished Reliability and Established Reliability. These nonestablished and established reliability multiturn lead screw actuated, and single-turn nonwire wound variable resistors with a contact bearing uniformly over the surface of the entire resistive element, when position by an actuator. These resistors are capable of full load operation (when the maximum resistance is engaged), at maximum ambient temperature of +85°C and are suitable for continuous operation, when properly derated, at a maximum ambient temperature of +150°C. Use for matching, balancing, and adjusting circuit variables in computers, telemetering equipment, and other critical applications. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-PRF-22097 inactive specification sheets.

TABLE A-V. Variable nonwire wound resistor selection guidance table.

Specification	Styles available	Power and max voltage ratings	Resistance tolerance (± percent) (see A.1.1.2)	Resistance Range (ohms)	Maximum body size (inches) and configuration (see figure A-2)
MIL-PRF-94 Composition	RV2 RV4 RV5 RV6 2RV7 RV8	1.0/0.5* 2.0/1.0* 0.5/0.25* 0.5/0.25* various 0.5/0.25*	±10/±20 ±10/±20 ±10/±20 ±10/±20 ±10/±20 ±10/±20	100 to 2.5M 50 to 2.5M 250 to 2.5M 100 to 5M 50 to 5M 100 to 5M	0.453 x 0.906 0.609 x 1.094 0.375 x 0.750 0.453 x 0.500 1.266 x 1.094 0.581 x 0.500 F
MIL-PRF-22097 Adjustment type	RJ12 RJ24	0.75 0.50	±10	10 to 1M	1.250 x 0.190 x 0.315 (I) 0.375 x 0.150 x 0.375 (H)
MIL-PRF-39023 Precision	RQ051 RQ090 RQ091 RQ100 RQ110 RQ150 RQ160 RQ200 RQ210 RQ300	0.50 1.00 1.00 2.50 1.25 1.50 3.50 2.00 4.50 3.00	±10	5000 100 to 1M 100 to 1M 1000 to 1M 100 to 1M 100 to 1M 1000 to 3M 100 to 1M 1000 to 3M 100 to 1M	0.9644 x 0.550 0.906 x 0.810 0.906 x 0.750 0.906 x 1.880 1.125 x 0.810 1.500 x 1.060 1.468 x 2.500 2.031 x 1.310 2.031 x 2.900 3.031 x 1.310 H
MIL-PRF-39035 Adjustment type	RJR24 RJR26 RJR50	0.50 0.25 0.25	±10	10 to 1M	0.375 x 0.150 x 0.375 (H) 0.270 x 0.165 x 0.250 (H) 0.25** x 0.25 (J)

* A taper/C and F taper
** Diameter

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APPENDIX A

A.3.1.6 Special Resistors. See table A-VI.

A.3.1.6.1 MIL-PRF-29, MF, Resistors, Fixed, Meter Multiplier, External (High Voltage, Ferrule Terminal Type). These resistors are used for high-voltage, external, meter-multiplier, fixed resistors of the ferrule-terminal type for use with direct-current (dc) instruments drawing 1 milliampere at full-scale deflection.

A.3.6.1.2 MIL-PRF-23648, RTH, Resistor, Thermal (Thermistor) Insulated. These resistors exhibit a rapid change in resistance for a relative small temperature change. These resistors are used to measure temperature or to compensate for changes in temperature.

A.3.1.6.3 MIL-PRF-32192, RCTP and RCTN, Resistor Chip, Thermal. These thermistors exhibit a positive temperature coefficient (PTC) or a negative temperature coefficient (NTC) and are primarily intended for incorporation into surface mount applications. These devices are to be used for temperature control, temperature compensation, sensing, and frequency compensation over the temperature range specified.

A.3.1.6.4 MIL-PRF-83530, RVS, Resistors, Voltage Sensitive Resistor, (Varistor, Metal-Oxide). These devices function as a nonlinear variable impedance dependent on voltage. They are designed to protect a circuit from a surge in voltage.

TABLE A-VI. Special resistors selection guidance table.

Specification	Styles available	Resistance range (in megohms)	Voltage ratings	Maximum body size (inches)
MIL-PRF-29 Resistors, Fixed, Meter Multiplier, External (High Voltage, Ferrule Terminal Type)	MFA	3.5, 4.0, 5.0, 6.0	3.5, 4.0, 5.0, 6.0	9.781 x 1.312
	MFB	1.0, 1.5, 2.0, 2.5, 3.0, 3.5	1.0, 1.5, 2.0, 2.5, 3.0, 3.5	5.281 x 1.312
	MFC	0.5, 0.8, 1.0	0.5, 0.8, 1.0	2.937 x 1.000
	MFD	10	10	16.500 x 1.312
	MFE	15	15	23.500 x 1.312
	MFF	20	20	30.500 x 1.312

Specification	Styles available	Power rating (Watts)	Thermal time constant (Seconds)	Dissipation constant mW/^{O}C	Resistance tolerance (percent) see A.1.1.2	Resistance Ratio 1/	Resistance range (ohms)	Maximum body size (inches) and configuration (see figure A.2)
MIL-PRF-23648 Resistor, thermal (thermistor) insulated	RTH06	0.05	80	5	F, G, J, K	A, B, C	68 min 75k max	0.30 D. M
	RTH22	0.5	60	15	J, K	E	10 min 39k max	0.43 L. x 0.16 D. A
	RTH42	0.25	60	2.5			10 min 10k max	0.30 L. x 0.11 D. A
	RTH44	0.2	25	2.0	F, G, J, K	A, B, C	300 min 500k max	0.25 L. x 0.13 D. x 0.135 M

1/ A - 19.8, B - 29.4, C - 48.7, D - 0.5, E - 0.55

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TABLE A-VI. Special resistors selection guidance table - Continued.

Specification	Styles available	Power rating (Watts)	Thermal time constant (Seconds)	Dissipation constant mW/^{O}C	Resistance tolerance (percent) (see A.1.1.2)	Resistance ratio (ohms)	Maximum body size (inches) (L x W x H) and configuration (see figure A.2)
MIL-PRF-32192 Resistors, chip thermal	RCTP0303	.125	30	1.25	F, G, J, K	0.53	0.048 x 0.048 x .082 T
	RCTP0805	.25	30	2.5			0.086 x 0.057 x 0.055 T
	RCTN0404	.0625	10	.625		Variable (based on resistance)	0.050 x 0.050 x .020 T
	RCTN0805	.125	8	2.0			0.086 x 0.057 x 0.055 T
	RCTN1206	.250	8	2.0			0.134 x 0.068 x 0.055 T

Specification	PIN M83530/1	Nominal varistor voltage (V)	Voltage rating (V) rms	Voltage rating (V) dc	Energy rating (joules)	Clamping voltage at 100 A (V)	Capacitance at 1 MHz (pF)	Clamping voltage at peak current rating (V)	Max body size (inches) A x D x E configuration (see figure A.2)
MIL-PRF-83530 Resistors, voltage sensitive, (varistor, metal-oxide)	-2000B	200	130	175	50	325	3800	570	0.110 x 0.95 x 0.32 (M)
	-2200D	220	150	200	55	360	3200	650	
	-4300E	430	275	369	100	680	1800	1200	
	-5100E	510	320	420	120	810	1500	1450	

A.3.2 Mounting guide.

A.3.2.1 Stress mounting. Improper heat dissipation is the predominant contributing cause of failure for any resistor type; consequently, the lowest possible resistor surface temperature should be maintained. Figure A-3 illustrates the manner in which heat is dissipated from fixed resistors in free air. The intensity of radiated heat varies inversely with the square of the distance from the resistor. Maintaining maximum distance between heat generating components serves to reduces cross radiation heating effects and promotes better convection by increasing air flow. For optimum cooling without a heat sink, small resistors should have large diameter leads of minimum length terminating in tiepoints of sufficient mass to act as heat sinks. All resistors have a maximum surface temperature which should never be exceed. Any temperature beyond maximum can cause the resistor to malfunction. Resistors should be mounted so that there are no abnormal hot spots on the resistor surface. When mounted, resistor should not come in contact with heat insulating surfaces.

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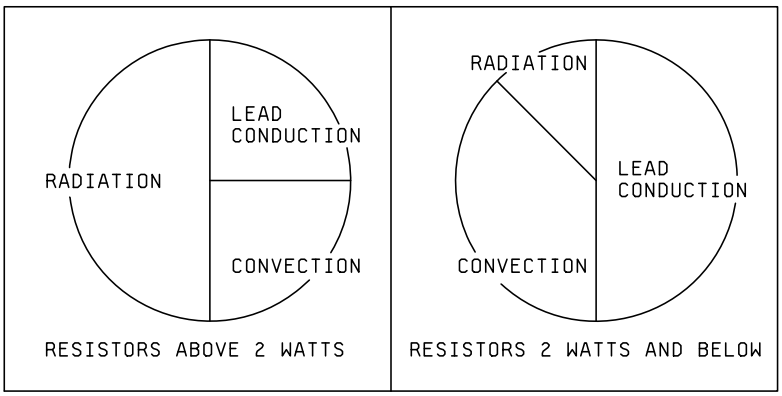


FIGURE A-3. Heat dissipation of resistors under room conditions.

A.3.2.2 Resistor mounting for vibration. Resistors should be mounted so resonance does not occur within the frequency spectrum of the vibration environment to which the resistors may be subjected. Some of the most common resistor packaging methods result in large resistor noise. Resistor mounting for vibration should provide (1) the least tension or compression between the lead and body, (2) the least excitation of the resistor in relation with any other surface, and (3) no bending or distortion of the resistor body.

A.3.2.3 Circuit packaging. Resistors that are crowed together and come into contact with each other can provide leakage paths (even well insulated parts) for external current passage. This can change the resultant resistance in the circuit. Moisture traps and dirt traps are easily formed by crowding. Moisture and dirt eventually form corrosive materials which can deteriorate the resistors and other electronic parts. Moisture can accumulate around dirt even in an atmosphere of normal humidity. Planning should be done to eliminate crowding of parts. Proper space utilization of electronic parts can reduce the package size and still provide adequate spacing of parts.

A.3.2.4 Summary. The following is a guide for resistor mounting:

a. Maintain lead length to a minimum. The mass of the point acts as a heat sink. (NOTE: Where low temperatures are present, leads should be offset (bent slightly) to allow for thermal contraction).

b. Close tolerance and low value resistors require special precautions (such as short leads and good soldering techniques) since the resistance of the leads and the wiring may be as much as several percent of the resistance of the resistor.

c. Maintain maximum spacing between resistors.

d. For resistors mounted in series, consider the heat being conducted through the leads to the next resistor.

e. Large power units should be mounted to the chassis.

f. Do not mount high power units directly on terminal boards or printed circuits.

g. To provide for the most efficient operation and even heat distribution, power resistors should be mounted in a horizontal position.

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h. Select mounting materials that will not char and can withstand due to expansion.

i. Consider proximity to other heat sources as well as self-heat.

j. Consider levels of shock and vibration to be encountered. Where large body mass is present, the body should be restrained from movement.

A.3.3 Effects of circuit usage. Resistors must be selected to be compatible with the conditions to which they are exposed. Numerous matters must be considered in this selection process. The most important are noted in the following.

A.3.3.1 Resistance value. This is initially determined by the circuit requirements, and may seem a trivial thing to mention. However, most resistor calculations that are made without reference to available resistors come out to a resistance value that are not standard. The design engineer should be aware of the standard resistance values that are available from manufacturers who adhere to this handbook and various Department of Defense specifications for resistors. These differ somewhat with the various types of resistors. It is a usually a fairly simple thing to bring the exact calculated value in line with a standard value. In the case where this cannot be done, a parallel or series combination of resistors can usually be used. The design engineer should also remember that the resistance value of the resistor that gets into the physical circuit will differ from the value that is stated on the circuit schematic, and that this difference will change as time goes by. The purchase tolerance of the resistor to be used will allow it to differ from the nominal stated value, depending on the type of resistor specified. Furthermore, the temperature at which the resistor works, the voltage across it, and the environment in which it lives will affects the actual value at particular times. For example, the designer should allow for a possible variation of ±15 percent from the nominal value of a purchased ±5 percent composition resistor, for the circuit to continue to operate satisfactorily over a very long time under moderate ambient conditions. Such a figure is a rule of thumb, based on many tests, and many resistors will remain much nearer their starting value; but if many are used, chance will ensure that some will go near this limit. A similar figure can be deduced from each variety of resistor used.

A.3.3.1.1 Summary.

a. Select a resistor for each circuit application from the lists of standard types and values.

b. Be sure that the circuit being designed will work with any resistor whose resistance value is within the limits set by tolerance plus voltage coefficient plus temperature coefficient plus drift with time. Failure to take these precautions can possibly mean that in equipment produced in quantity for the armed services, there may be some circuits that will not work under extreme conditions.

c. Various initial tolerances are available depending on the type of resistor. It should be remembered that initial accuracies become meaningless if the inherent stability of the resistor does not support the initial accuracy.

d. During shelf life, as well as during operational life, any characteristic (such as resistance, power rating, dielectric strength, or size) of any part may change value due to stresses caused by environmental changes of temperature , humidity, pressure or vibration. Changes of characteristic caused by environmental stresses may be linear or nonlinear, reversible or nonreversible (permanent), or combinations thereof. Where a characteristic of the part undergoes a linear change during environmental stress, and the change reverses itself linearly when the environmental stress is removed so that characteristic returns to its normal value, this rate of change in characteristic value (per unit change in stress value) is designated (x) coefficient, and is usually expressed in percent or ppm/^{O}C.

A.3.3.2 Power rating. The minimum required power rating of a resistor is another factor that is initially set by the circuit usage, but is markedly affected by other conditions of use. As mentioned previously, the power rating is based on the hot spot temperature the resistor will withstand, while still meeting its other requirements of resistance variation, accuracy, and life. For derating information see A.3.3.2.2.

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A.3.3.2.1 Self-generated heat. Self generated-heat in a resistor is, of course, calculated as $P = I^2R$. This figure, in any circuit, must be less than the actual power rating of the resistor used. It is the usual practice to calculate this value and to use the next larger power rating available in the handbook. This calculation should, however, be considered only as a first approximation of the actual rating to be used.

A.3.3.2.2 Rating versus ambient conditions. The power rating of a resistor is based on a certain temperature rise from an ambient temperature of a certain value. If the ambient temperature is greater than this value, the amount of heat that the resistor can dissipate is correspondingly reduced, and therefore it must be derated because of temperature. The applicable section of this handbook and all of the Department of Defense specifications contain derating curves to be applied to the resistors covered.

A.3.3.2.3 Rating versus accuracy. Because of temperature coefficient of resistance that all resistors possess, a resistor which is expected to remain near its measured value under conditions of operation must remain cool. For this reason, all resistors designated as “accurate” are very much larger physically for a certain power rating than are ordinary “nonaccurate” resistors. In general, any resistor, “accurate” or not, must be derated to remain very near its original measured value when it is being operated.

A.3.3.2.4 Rating versus life. If especially long life is required of a resistor, particularly when “life” means remaining within a certain limit of resistance drift, it is usually necessary to derate the resistor, even if ambient conditions are moderate and if accuracy by itself is not important. A good rule to follow when choosing a resistor size for equipment that must operate for many thousands of hours is to derate it to one half of its nominal power rating. Thus, if the self-generated heat in the resistor is one third watt, do not use a one half watt resistor, but rather 1 watt size. This will automatically keep the resistor cooler, will reduce the long term drift, and will reduce the effect of the temperature coefficient. In equipment that need not live so long and must be small in size, this rule may be impractical, and the engineer should adjust the dependence on the rules to the circumstances at hand. A “cool” resistor will generally last longer than a “hot” one, and can absorb transient overloads that might permanently damage a “hot” resistor.

A.3.3.2.5 Rating under pulsed conditions and under intermittent loads. When a resistor is used in circuits where power is drawn intermittently or in pulses, the actual power dissipated with safety during the pulses can sometimes be much more than the maximum rating of the resistor. For short pulses, the actual heating is determined by the duty factor and the peak power dissipated. Before approving such a resistor application, however, the engineer should be sure (1) that the maximum voltage applied to the resistor during the pulses is never greater than the permissible maximum voltage for the resistor be used, (2) that the circuit cannot fail in such a way that continuous excessive power can be through the resistor and cause it to fail also, (3) that the average power being drawn is well within the agreed on rating of the resistor, and (4) that continuous steep wavefronts applied to the resistor do not cause any unexpected troubles.

A.3.3.3 High frequency. For most resistors, the lower the resistance value, the less total impedance it exhibits at high frequency. Resistors are not generally tested for total impedance at frequencies above 120 Hz. Therefore, this characteristic is not controlled. The dominating conditions for good high frequency resistor performance are geometric considerations and minimum dielectric losses. For the best high frequency performance, the ratio of resistor length to the cross sectional area should be a maximum. Dielectric losses are kept low by proper choice of the resistor base material, and when dielectric binders are used, their total mass is kept to a minimum. The following is a discussion of the high frequency merits of these major resistor types:

a. Film type. Film type resistors have the best high frequency performance. The effective dc resistance for most resistance values remains fairly constant up to 100 MHz and decreases at higher frequencies. In general, the higher the resistance value the greater the effect of frequency.

b. Wire wound. Wire wound resistors have inductive and capacitive effects and are unsuited for use above 50kHz, even when specially wound to reduce the inductance and capacitance. Wire wound resistors usually exhibit an increase in resistance with high frequencies because of “skin” effect.

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A.3.4 Effects of mechanical design and ambient conditions. Since the operation of a circuit cannot be divorced from the physical configuration it assumes when assembled, some of the points that apply herein have already been discussed. It is well, however, to check this aspect of equipment design several times, so redundancies in the following paragraphs are deliberate for the sake of emphasis.

A.3.4.1 Mechanical design of resistors. Much trouble during the life of the equipment can be eliminated if the design engineer can be sure that the resistors he is specifying for his circuits are soundly constructed and proper equipment assembly techniques are utilized. The resistor types listed in this handbook provide a great measure of this assurance and, in general, assure a uniform quality of workmanship. The areas defined in 3.4.1.1 through 3.4.1.7 are included as indicators of sound product construction.

A.3.4.1.1 End caps or terminations. The connection between the resistor element itself and the pigtails or leads that connect it into the circuit must be so good that no possible combination of conditions met in the proposed service can cause an intermittent connection. The Department of Defense specifications cover this point, and provide tests to check for it. When resistors are handled in automatic assembly machines, this precaution is particular important.

A.3.4.1.2 Effect of soldering. There are assembly techniques that affect resistor reliability. Resistors should never be overheated by excessive soldering iron applications, and the resistor leads should not be abraded by assembly tools. No normal soldering practice, either manual or dip soldering, should damage the resistor physically or change its resistance value appreciably.

A.3.4.1.3 Moisture resistance. Moisture is the greatest enemy of components and electronic equipment. Usually a resistor will keep itself dry because of its own self-generated heat; this is, of course, only true when the equipment is turned on. If the equipment must stand for long periods under humid conditions without power applied, the engineer should determine whether his circuits will operate with resistance values which have changed from the “hot” condition, and whether the retrace of the resistance value during the warm up period will allow the equipment to work satisfactorily during this period. If it will not, the resistor must be adequately protected against moisture absorption. The resistor cannot be blamed for performing improperly if it is not designed for the use to which it is put. It is therefore up to the design engineer to analyze what is needed and to provide the resistor to meet these conditions. This handbook and the applicable Department of Defense specifications constitute a guide as to what various kinds of resistors will do under humid conditions.

A.3.4.1.4 Method of mounting. Large resistors that are not provided with adequate means of mounting should not be considered, under conditions of vibration or shock. Lead failure can occur, and the larger the mass supported by the leads the more probable a failure will be. Even when vibration or shock will not be a serious problem, ease of assembly and replaceability suggest that large components be mounted individually.

A.3.4.1.5 Resistor body. The body of the resistor must be sufficiently strong to withstand any handling it is likely to get. The specifications call out, through workmanship and packaging requirements, that it be shown by the manufacturer that the product will not crack, chip, or break in transit, on the shelf, or in the normal assembly process.

A.3.4.1.6 Insulation or coating. All resistors intended for use in reliable electronic equipment must be protected by an insulating coating. Sometimes this is a molded phenolic case, epoxy coating, or ceramic or glass sleeves. Wire wound power resistors use various cement and vitreous enamel coatings to protect the windings, and to insulate and provide moisture barriers. Not all of the coatings and insulations applied to commercial resistors are satisfactory for extreme variations in ambient conditions; the various Department of Defense specifications include test used to qualify the various manufacturer’s products thus providing a greater confidence in the coating used.

A.3.4.1.7 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of resistor components and solder will not exceed 97 percent, by mass. Tin will be alloyed with a minimum of 3 percent lead, by mass.

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A.3.4.2 Effects of ambient conditions. In the establishment of rating for resistors, the design engineer has implicitly considered the mechanical design of the equipment. This may not have been realized, but it is so because the ambient conditions in which the resistor must operate determine to a large degree the power rating and mechanical construction of the resistor if long life, or any life, under extreme conditions is desired.

A.3.4.2.1 Resistor heating. A very important question in the application of resistors is how hot will they get in service. In a piece of equipment the heat in a resistor comes from several sources; namely, (1) self-generated heat, and is the thing that can be easily calculated, and (2) the heat that the resistor receives from other resistors or other heat producing components in the same immediate neighborhood by radiation, and is not so easily calculated. The important thing to remember is under these conditions each resistor will be heated more than I^2R would suggest; when much heat is produced, as in stacked wire wound resistors, the design engineer would do well not to freeze his design until he has measured a typical assembly with power on to see just how hot the resistor gets. The same thing is true of the extra heating given the resistors by convection. This is another way of saying that high ambient temperature will reduce the actual power rating of the resistor by reducing permissible temperature rise, a point that has been made several times before. The equipment designer must realize also that the heat being produced by "hot" resistors can injure other components. This is a very important point to remember; capacitors, diodes, and other resistors usually do not fail immediately when overheated. The effect of too much heat is a deteriorating one, weakening the component until at a later date it will unexpectedly fail. It is very easy to put a "heat bomb" in a piece of equipment that will not go off in normal production testing but will do its duty. It is also very easy to eliminate such troubles by strict and thoughtful attention to the problem of heating. A few rules have been given for use as guides to protect against these factors. (See A.3.3.2)

A.3.4.2.2 High altitude. With the exception of the dielectric withstanding voltage test at reduced barometric pressure, all tests in Department of Defense specifications referenced herein are performed at ambient atmospheric pressure. This fact should be considered when the use of these resistors for high altitude conditions is contemplated.

A.3.4.2.3 Flammability. It should be noted that Department of Defense specifications referenced herein contain no requirements concerning the flammability of the materials used in the construction of these resistors. Users should take this into consideration when a particular application involves this requirement.

A.4. SUPPLEMENTAL INFORMATION

A.4.1 Reliability. The established reliability specification provides for the establishment of a failure rate figure through the single parameter of load life only. Although, in most instances, the established reliability specification provides for more frequent moisture resistance, burn in, and other types of screening tests on a 100 percent basis, the failure rate (in percent per 1,000 hours) is based only on load life test results.

A.4.2 Metric equivalent. The metric equivalents (to the nearest 0.01 mm) which are provided in the individual sections are for general information only and are based upon 1 inch = 25.4 mm.

A.4.3 International standardization agreements. Certain provisions of the specifications referenced in this handbook are subjected on international standardization agreements. When amendment, revision, or cancellation of any of these specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

A.4.4 Cross reference. A cross reference of section number, Department of Defense specification numbers, associated specification numbers, and style numbers are included for reference (see table A-VII).

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APPENDIX A

A.4.5 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

TABLE A-VII. Detailed specification by style number.

DoD specification	Style
MIL-PRF-19/2	RA20
MIL-PRF-19/3	RA30
MIL-PRF-22/1	RP06
MIL-PRF-22/2	RP07
MIL-PRF-22/3	RP10
MIL-PRF-22/4	RP11
MIL-PRF-22/5	RP15
MIL-PRF-22/6	RP16
MIL-PRF-22/7	RP20
MIL-PRF-22/8	RP25
MIL-PRF-22/9	RP30
MIL-PRF-22/10	RP35
MIL-PRF-22/11	RP40
MIL-PRF-22/12	RP45
MIL-PRF-22/13	RP50
MIL-PRF-22/14	RP55
MIL-PRF-22/15	RP05
MIL-PRF-26/3	RW29
MIL-PRF-26/3	RW31
MIL-PRF-26/3	RW33
MIL-PRF-26/3	RW35
MIL-PRF-26/3	RW37
MIL-PRF-26/3	RW38
MIL-PRF-26/3	RW47
MIL-PRF-26/4	RW56
MIL-PRF-29	MFA
	MFB
	MFC
	MFD
	MFE
	MFF
MIL-PRF-94/2	RV5
MIL-PRF-94/3	RV6
MIL-PRF-94/4	RV2
MIL-PRF-94/5	RV4
MIL-PRF-94/6	2RV7
MIL-PRF-94/7	RV8
MIL-PRF-914/3	RNS030
MIL-PRF-914/4	RNS040
MIL-PRF-914/5	RNS050

DoD specification	Style
MIL-PRF-12934/1	RR0900
MIL-PRF-12934/2	RR1100
MIL-PRF-12934/4	RR2000
MIL-PRF-12934/5	RR3000
MIL-PRF-12934/6	RR1000
MIL-PRF-12934/9	RR2100
MIL-PRF-12934/10	RR3100
MIL-PRF-12934/15	RR3200
MIL-PRF-12934/16	RR3300
MIL-PRF-12934/17	RR3400
MIL-PRF-12934/18	RR3500
MIL-PRF-12934/19	RR1300
MIL-PRF-12934/20	RR1400
MIL-PRF-12934/27	RR3600
MIL-PRF-12934/28	RR3700
MIL-PRF-12934/29	RR3800
MIL-PRF-12934/30	RR3900
MIL-PRF-12934/31	RR4000
MIL-PRF-12934/32	RR4100
MIL-PRF-12934/33	RR2002
MIL-PRF-12934/34	RR1004
MIL-PRF-12934/35	2RR2104
MIL-PRF-12934/36	RR3601
MIL-PRF-12934/37	2RR3100
MIL-PRF-18546/2	RE77
MIL-PRF-18546/2	RE80
MIL-PRF-22097/2	RJ12
MIL-PRF-22097/4	RJ24
MIL-PRF-22684/8	RL42...TX
MIL-PRF-23648/1	RTH06
MIL-PRF-23648/9	RTH22
MIL-PRF-23648/19	RTH42
MIL-PRF-23648/20	RTH44
MIL-PRF-27208/8	RT12
MIL-PRF-27208/9	RT24
MIL-PRF-27208/10	RT26

DoD specification	Style
MIL-PRF-32192/1	RCTP0303
MIL-PRF-32192/2	RCTP0805
MIL-PRF-32192/3	RCTN0404
MIL-PRF-32192/4	RCTN0805
MIL-PRF-32192/5	RCTN1206
MIL-PRF-32159/1	RCZ0502
MIL-PRF-32159/2	RCZ0505
MIL-PRF-32159/3	RCZ1005
MIL-PRF-32159/4	RCZ1505
MIL-PRF-32159/5	RCZ2208
MIL-PRF-32159/6	RCZ0705
MIL-PRF-32159/7	RCZ1206
MIL-PRF-32159/8	RCZ2010
MIL-PRF-32159/9	RCZ2512
MIL-PRF-32159/10	RCZ1010
MIL-PRF-32159/11	RCZ0402
MIL-PRF-32159/12	RCZ0603
MIL-PRF-32159/13	RCZ0302
MIL-PRF-39002/1	RK09
MIL-PRF-39002/3	RK11
MIL-PRF-39005/1	RBR52
MIL-PRF-39005/2	RBR53
MIL-PRF-39005/3	RBR54
MIL-PRF-39005/4	RBR55
MIL-PRF-39005/5	RBR56
MIL-PRF-39005/6	RBR71
MIL-PRF-39005/11	RBR80
MIL-PRF-39005/11	RBR81
MIL-PRF-39007/7	RWR78
MIL-PRF-39007/8	RWR80
MIL-PRF-39007/9	RWR81
MIL-PRF-39007/10	RWR84
MIL-PRF-39007/11	RWR89
MIL-PRF-39007/12	RWR82

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TABLE A-VII. Detailed specification number by style number - Continued.

DoD specification	Style
MIL-PRF-39009/1	RER60
MIL-PRF-39009/1	RER65
MIL-PRF-39009/1	RER70
MIL-PRF-39009/1	RER75
MIL-PRF-39009/2	RER40
MIL-PRF-39009/2	RER45
MIL-PRF-39009/2	RER50
MIL-PRF-39009/2	RER55
MIL-PRF-39015/1	RTR12
MIL-PRF-39015/2	RTR22
MIL-PRF-39015/3	RTR24
MIL-PRF-39017/1	RLR07
MIL-PRF-39017/2	RLR20
MIL-PRF-39017/3	RLR32
MIL-PRF-39017/5	RLR05
MIL-PRF-39023/1	RQ090
MIL-PRF-39023/2	RQ110
MIL-PRF-39023/3	RQ150
MIL-PRF-39023/4	RQ200
MIL-PRF-39023/5	RQ300
MIL-PRF-39023/6	RQ100
MIL-PRF-39023/7	RQ160
MIL-PRF-39023/8	RQ210
MIL-PRF-39023/9	RQ091
MIL-PRF-39023/10	RQ051
MIL-PRF-39035/2	RJR24
MIL-PRF-39035/3	RJR26
MIL-PRF-39035/4	RJR50

DoD Specification	Style
MIL-PRF-49462/3	RHV30
	RHV31
	RHV32
	RHV33
	RHV34
	RHV35
MIL-PRF-49465/1	RLV10
MIL-PRF-49465/6	RLV30
MIL-PRF-49465/7	RLV31
MIL-PRF-55182/1	RNC/N/R55
MIL-PRF-55182/3	RNC/N/R60
MIL-PRF-55182/5	RNC/N/R65
MIL-PRF-55182/6	RNC/N/R70
MIL-PRF-55182/7	RNC/N/R50
MIL-PRF-55182/9	RNC/N90
MIL-PRF-55182/10	RNC/N/R75
MIL-PRF-55342/1	RM0502
MIL-PRF-55342/2	RM0505
MIL-PRF-55342/3	RM1005
MIL-PRF-55342/4	RM1505
MIL-PRF-55342/5	RM2208
MIL-PRF-55342/6	RM0705
MIL-PRF-55342/7	RM1206
MIL-PRF-55342/8	RM2010
MIL-PRF-55342/9	RM2512
MIL-PRF-55342/10	RM1010
MIL-PRF-55342/11	RM0402
MIL-PRF-55342/12	RM0603
MIL-PRF-55342/13	RM0302

DoD Specification	Style
MIL-PRF-83401/1	RZ010
MIL-PRF-83401/2	RZ020
MIL-PRF-83401/3	RZ030
MIL-PRF-83401/4	RZ040
MIL-PRF-83401/5	RZ050
MIL-PRF-83401/6	RZ060
MIL-PRF-83401/7	RZ070
MIL-PRF-83401/8	RZ080
MIL-PRF-83401/9	RZ090
MIL-PRF-83401/10	RZ100
MIL-PRF-83401/13	RZ130
MIL-PRF-83401/14	RZ140
MIL-PRF-83401/15	RZ150
MIL-PRF-83401/18	RZ180
MIL-PRF-83401/19	RZ190
MIL-PRF-83401/21	RZ210
MIL-PRF-83401/22	RZ220
MIL-PRF-83401/23	RZ230
MIL-PRF-83401/24	RZ240
MIL-PRF-83530/1	RVS10

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APPENDIX B

COMMERICAL ITEM DESCRIPTIONS (CIDs)

B.1. SCOPE

B.1.1 Scope. The following appendix includes all active Commercial Item Descriptions (CIDs). These drawings are used to supplement Military Specifications.

B.2 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

B.3 COMMERICAL ITEM DESCRIPTIONS (CIDs)

B.3.1 Commercial Item Descriptions (CIDs). Table D-I show a listing of Commercial Item Descriptions (CIDs) available for use.

B-I. List of Commercial Item Descriptions (CIDs).

CID Number	Title
A-A-55088	Resistors, Flameproof, Fusible
A-A-55501	Resistors, Fixed, Zero Ohm, 1/4 Watt
A-A-55502	Resistors, Fixed, Zero Ohm, 1/8 Watt
A-A-55512	Resistor, Thermally Sensitive (Thermistor)
A-A-55517	Resistor, Fixed, Carbon Film, General Requirements For
A-A-55517/1	Resistor, Fixed, Carbon Film, 1/8 Watt
A-A-55517/2	Resistor, Fixed, Carbon Film, 1/4 Watt
A-A-55517/3	Resistor, Fixed, Carbon Film, 1/2 Watt
A-A-55517/4	Resistor, Fixed, Carbon Film, 1 Watt
A-A-55517/5	Resistor, Fixed, Carbon Film, 2 Watt
A-A-55534	Resistor, Fixed, Power Type (Very Low Resistance Values), General Requirements For
A-A-55534/1	Resistors, Fixed, Wire Wound or Metal Element, (Power Type) Style VLV1
A-A-55534/2	Resistors, Fixed, Wire Wound or Metal Element, (Power Type) Style VLV1206
A-A-55534/3	Resistors, Fixed, Wire Wound or Metal Element, (Power Type) Style VLV3
A-A-55534/4	Resistors, Fixed, Wire Wound or Metal Element, (Power Type) Style VLV5
A-A-55534/5	Resistors, Fixed, Wire Wound or Metal Element, (Power Type) Style VLV7
A-A-55534/6	Resistors, Fixed, Wire Wound or Metal Element, (Power Type) Style VLV10
A-A-55534/7	Resistors, Fixed, Wire Wound or Metal Element, (Power Type) Style VLV2010
A-A-55534/8	Resistors, Fixed, Wire Wound or Metal Element, (Power Type) Style VLV2512
A-A-55534/9	Resistors, Fixed, Wire Wound or Metal Element, (Power Type) Style VLV2
A-A-55562	Resistor, Voltage Sensitive (Varistor, Metal Oxide), Chip
A-A-55562/1	Resistor, Voltage Sensitive (Varistor, Metal Oxide), Chip, Style 0603
A-A-55562/2	Resistor, Voltage Sensitive (Varistor, Metal Oxide), Chip, Style 0805
A-A-55562/3	Resistor, Voltage Sensitive (Varistor, Metal Oxide), Chip, Style 1206
A-A-55562/4	Resistor, Voltage Sensitive (Varistor, Metal Oxide), Chip, Style 1210
A-A-55562/5	Resistor, Voltage Sensitive (Varistor, Metal Oxide), Chip, Style 0402

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B-I. List of Commercial Item Descriptions (CIDs). - Continued

CID Number	Title
A-A-55564	Resistor, Voltage Sensitive (Varistor), General Specification for
A-A-55564/1	Resistor, Voltage Sensitive (Varistor), Base Mount
A-A-55564/2	Resistor, Voltage Sensitive (Varistor), Metal Oxide, High Energy
A-A-55564/3	Resistor, Voltage Sensitive (Varistor), Metal Oxide, Radial Lead
A-A-59496	Resistor, Variable, Trimmer Nonwirewound, and Wirewound General Requirements for
A-A-59496/1	Resistor, Variable, Trimmer Nonwirewound, 4 mm Square, J-Hook Leads
A-A-59496/2	Resistor, Variable, Trimmer Nonwirewound, 4 mm Square, Gull Wing Leads
A-A-59497	Resistors, Variable, Nonwirewound, Wire Wound, Precision, Nonprecision, General Requirements for
A-A-59497/1	Resistors, Variable, Nonwirewound, Nonprecision, 1 Watt
A-A-59497/2	Resistors, Variable, Nonwirewound, Precision, Linear Motion Position Sensor
A-A-59714	Resistors, Resettable Fuses (Non-Linear Thermistor) General Requirements for
A-A-59714/1	Resistor, Resettable Fuses (Non-Linear Thermistor) Radial Leaded, Style RXE
A-A-59714/2	Resistor, Resettable Fuses (Non-Linear Thermistor) Radial Leaded, Style RUE
A-A-59714/3	Resistor, Resettable Fuses (Non-Linear Thermistor) Radial Leaded, Style TR
A-A-59715	Resistors, Fixed and Adjustable, Wire Wound, Power Type, General Requirements for
A-A-59715/1	Resistors, Fixed, Wire Wound, Power Type, Lead Terminal
A-A-59715/2	Resistors, Adjustable, Wire Wound, Power Type
A-A-59715/3	Resistors, Fixed, Wire Wound, Power Type
A-A-59715/4	Resistors, Fixed, Wire Wound, Power Type, Tab Terminals
A-A-59715/5	Resistors, Fixed and Adjustable, Wire Wound, Power Type
A-A-59769	Resistor Networks, Ball Grid Array Terminators General Requirements for
A-A-59769/1	Resistor Networks, Ball Grid Array Terminators Thevenin Termination

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APPENDIX C

DSCC DRAWINGS

C.1. SCOPE

C.1.1 Scope. The following appendix includes all active Defense Supply Center, Columbus (DSCC) Drawings. These drawings are used to supplement Military Specifications.

C.2 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

C.3 DSCC DRAWINGS

C.3.1 DSCC Drawings. Table C-I show a listing of DSCC Drawings available for use.

C-I. List of DSCC Drawings

Drawing Number	Title
00001	Resistor, Variable, Wirewound, Nonprecision, Trimmer, 3/8 Inch, Square, 1 Watt, Flexible Leads
01002	Resistor, Fixed, Chip, 1.5 Watt (MELF - 2512), Flat Ceramic Package
01032	Resistor, Fixed, Film, Chip, Beryllia Substrate, High Power, Style 1206
01033	Resistor, Fixed, Film, Chip, Voltage Divider, Style 1206
02001	Resistor, Fixed, Film, Precision, Chip 1/8 Watt, Style 2012
02005	Resistors, Variable, Nonwire Wound (Adjustment Type, Lead-Screw Actuated) 1/2" Rectangular, 0.3 Watts
02008	Resistor, Fixed, Film, Chip, Low and High Values, Style 1206
02010	Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 1206
03003	Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/8 Watt
03004	Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/4 Watt
03005	Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/2 Watt
03006	Resistor, Fixed, Carbon Film, High Pulse Voltage, 1 Watt
03007	Resistor, Fixed, Carbon Film, High Pulse Voltage, 3/4 Watt
03008	Resistor, Fixed, Carbon Film, High Pulse Voltage, 2 Watt
03009	Resistor, Fixed, Carbon Film, High Pulse Voltage, 3 Watt
03010	Resistor, Fixed, Film, Chip, Surface Mounted, Ultra Precision, Style 1506
03011	Resistor, Fixed, Film, Chip, Zero-Ohm, Style 0201
03022	Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0603
03025	Resistor, Chip, Fixed, Film, High Voltage, Style 1206
03026	Resistor, Chip, Fixed, Film, High Voltage, Style 2010
03027	Resistor, Chip, Fixed, Film, High Voltage, Style 2512
04007	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0302
04008	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0402
04009	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0603
04025	Resistor, Chip, Fixed, Film, Surface Mount, 5 Watts (Up to 25 Watts with Heatsink)
04032	Resistor, Fixed, Film, Chip, Low Values, High Power, Style 2512
05009	Resistor, Fixed, Film, Chip (MELF), 1/4 Watt, Style 0204
05015	Resistor, Chip, Fixed, Film, Surface Mount, 2.25 Watts (Up to 20 Watts with Heatsink)
05016	Resistor, Fixed, Film, Radial Lead, 2.25 Watts (Up to 20 Watts with Heatsink)
06001	Resistors, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2010

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APPENDIX C

C-I. List of DSCC Drawings (continued)

Drawing Number	Title
06002	Resistors, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2512
06003	Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (2 Watt)
06006	Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (3 Watt), Style 4527
06007	Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (.1 Watt), Style 0603
06008	Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (.125 Watt), Style 0805
06009	Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (2.0 Watt), Style 2816
06010	Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (.5 Watt), Style 2010
06011	Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (1.0 Watt), Style 2512
06012	Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (2.0 Watt), Style 2816
06018	Resistor Network, Fixed, Film, Surface Mount, Voltage Divider, 3 Pin
06020	Resistors, Fixed, Bulk Metal Foil, High Precision, Surface-Mount, Molded .25/.16 Watt
06021	Resistors, Fixed, Bulk Metal Foil, High Precision, Surface Mount, Molded 0.6/0.4 Watts
07002	Resistor, Fixed, Wirewound, Surface Mount, Power Type (3 Watt)
07005	Resistor, Chip, Fixed, Film, 8 Pin Array, Style 1206
07009	*placeholder*
07010	*placeholder*
07011	Resistors, Fixed, Current Sensing, Metal Strip, High Precision, Surface Mount, Style 2512
07012	Resistors, Fixed, Current Sensing, Metal Strip, High Precision, Surface Mount, Style 3637
07017	*placeholder*
07018	*placeholder*
07019	*placeholder*
07024	Resistors, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 0805
07025	Resistors, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 1206
08003	Resistors, Chip, Fixed, Current Sensing, Ultra Precision, Style 1625
85085	Resistor, Variable, Nonwire-Wound, Adjustment Type
86097	Resistor, Variable, Panel Mounted
87012	Resistor Network, Fixed, Film, Surface Mount, Gull Wing, 16 Pin
87013	Resistor Network, Fixed, Film, Surface Mount, Gull Wing, 14 Pin
87014	Resistor Network, 16 Pin, Leadless Chip Carrier
87015	Resistor Network, 28-Pin, Leadless Chip Carrier
87016	Resistor Network, 20-Pin, Leadless Chip Carrier
87017	Resistor Networks, Fixed, Film, Surface Mount, 20 Pin Leadless Chip Carrier
87018	Resistor Network, 16-Pin Leadless Chip Carrier
87025	Resistor Network, 8 Pin, Dual-in-Line Package (DIP)
87026	Resistor Network, Fixed, Film, 3-Pin
87027	Resistor Network, Fixed, Film, 3-Pin
87030	Resistor, Network, 6-Pin, Single-in-Line Package (SIP)
87031	Resistor, Network, 8-Pin, Single-in-Line Package (SIP)
87032	Resistor Network, 10-Pin, Single-in-Line Package (SIP)
87045	Resistor Network, 20-Pin, Hermetically Sealed, Dual-in-Line Package (DIP)

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APPENDIX C

C-I. List of DSCC Drawings (continued)

Drawing Number	Title
87053	Resistor, Network, 14-Pin, Flat Pack
87067	Resistor Network, Fixed, Film, 7-Pin Sip (Low Profile), Multiple Schematics
87068	Resistor Network, Fixed, Film, 7-Pin Sip (High Profile), Multiple Schematics
87071	Resistor Network, Fixed, Film, 10-Pin Sip, Multiple Resistance Values, Multiple Schematics, (Low Profile)
87072	Resistor Network, Fixed, Film, 10-Pin Sip, Multiple Resistance Values, Multiple Schematics and Multiple Tolerances.
87073	Resistor Network, Fixed, Film, Eight Pin SIP, Multiple Resistance Values, Multiple Schematics, (Low Profile)
87074	Resistor Network, Fixed, Film, Eight Pin SIP, Multiple Resistance Values, Multiple Schematics, (High Profile)
87075	Resistor, Fixed, Film, Chip, Flange Mount, Double Tab, High Power
87103	Resistor, Network, 9-Pin, SIP Multiple Values, Multiple Tolerances, (Low Profile)
87105	Resistor, Network, 9-Pin, SIP Multiple Values, Multiple Tolerances, (High Profile)
87126	Resistor, Variable, Nonwire-Wound (Adjustment Type, Lead Screw Actuated)
88008	Resistor, Tantalum Nitride, Chip
88009	Resistor, Fixed, Film, Chip, Tantalum Nitride
88014	Resistor, Network, 12-Pin, Single-in-Line Package (SIP)
88015	Resistor, Network, 8-Pin, Single Inline Package
88016	Resistor Network, 20 Pin Leadless Chip Carrier
88018	Resistor, Chip, Fixed, Film, Style 0705
88020	Resistor Network, 6-Pin, Leadless Chip Carrier
88027	Resistor, Chip, Fixed, Film, Style 0504
88030	Resistors, Chip, Fixed, Film, Style 1005
88031	Resistor, Chip, Fixed, Film, Style 1505
88034	Resistor, Fixed, Film, Chip, Low Values, High Power, Style 2512
88036	Resistor Network, 10 Pin, Leadless Chip Carrier
88040	Resistor, Variable, Surface Mount
89004	Resistor-Capacitor Network, 16 Pin DIP
89023	Resistor-Capacitor Network, 16 Pin Flat Pack
89039	Resistor, Fixed, Film, Precision
89040	Resistor, Fixed, Wire Wound, Surface Mount, Power Type (2.5 Watts)
89088	Resistor, Fixed, Film, Precision, 0.1 Watt, Power Curve C
89099	Resistor, Fixed, Film, Chip, Flange Mount, Single Tab, High Power
90038	Resistor, Fixed, Film, Precision, 0.25 Watt, Power Curve C
90039	Thermistor, Thermally Sensitive Resistor
92013	Resistor Network, 10-Pin, SIP, Extended Lead Length
92021	Resistor, Variable, Surface Mount, 4MM, Multiturn Trimmer, 1/4 Watt
93030	Resistors, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2018
93075	Resistor, Fixed, Wirewound, Surface Mount, Power Type (½ Watt)
93076	Resistor, Fixed, Wirewound, Surface Mount, Power Type (1 Watt)
93077	Resistor, Fixed, Wirewound, Surface Mount, Power Type (2 Watt)
93078	Resistors, Fixed, Bulk Metal Foil, Chip, Ultra Precision, Style 2110
93079	Resistors, Fixed, Bulk Metal Foil, Chip, Ultra Precision, Style 1505
94012	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0505

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APPENDIX C

C-I. List of DSCC Drawings (continued)

Drawing Number	Title
94013	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1005
94014	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 2208
94015	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0705
94016	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1206
94017	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 2010
94018	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 2512
94019	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1010
94025	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0502
94026	Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1505
94047	Resistor, Fixed, Chip, 2 Watt (MELF - 3610)
94048	Resistor, Fixed, Chip, 1/2 Watt (MELF - 2010)
95006	Resistor, Fixed, Chip, 1 Watt (MELF - 2512)
95011	Resistor, Fixed, Chip, 1/8 Watt (MELF - 1206)
95013	Resistor, Chip, Fixed, Film, Style 0302
95014	Resistor, Chip, Fixed, Film, Style 0402
96002	Resistor, Fixed, Film, Insulated, Low Inductance
97004	Resistors, Fixed, Film (Insulated), 2 Watt
97009	Resistor, Fixed, Film, Precision, 0.3 Watt, Power Curve A
97010	Resistor, Fixed, Film, Precision, 0.3 Watt, Power Curve B
97011	Resistor, Fixed, Film, Precision, 0.15 Watt, Power Curve B
97012	Resistor, Variable, Nonwire-Wound, Trimmer, 3/4 Watt, Flexible Leads
98020	Resistors, Fixed, Film (Insulated), 1/8 Watt
98021	Resistors, Fixed, Film (Insulated), 1/2 Watt
98022	Resistors, Fixed, Film (Insulated), 1 Watt
99001	Resistor, Fixed, Ceramic Composition (Insulated)
99004	Resistor, Fixed, Film, Chip, 4 Pin Array
99005	Resistor, Fixed, Film, Chip, 8 Pin Array
99006	Resistor, Fixed, Film, Chip, 10 Pin Array
99007	Resistor, Fixed, Film, Chip, 16 Pin Array
99011	Resistors, Fixed, Film (Insulated), 0.25 Watt

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APPENDIX D

INACTIVE FOR NEW DESIGN DoD SPECIFICATIONS, COMMERCIAL ITEM DESCRIPTIONS (CIDs) and DSCC DRAWINGS

D.1. SCOPE

D.1.1 Scope. The following appendix includes all inactive for new design DoD Specification, Commercial Item Descriptions (CIDs) and DSCC Drawings. These specifications are used for replacement purposes only.

D.2 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

D.3 INACTIVE DoD SPECIFICATIONS, COMMERCIAL ITEM DESCRIPTIONS (CIDs) AND DSCC DRAWINGS.

D.3.1 INACTIVE DoD SPECIFICATIONS.

D.3.1.1 Inactive DoD Specifications. Table D-I contains a listing of Inactive for new design DoD specifications.

D-I. Inactive DoD specifications.

Specification number	Title
MIL-R-26/2	Resistors, Fixed, Wire Wound (Power Type), Styles RW20, RW21, RW22, RW23, and RW24
MIL-R-26/5	Resistors, Fixed, Wire Wound (Power Type), Styles RW70, RW74, RW78, and RW79
MIL-R-26/6	Resistors, Fixed, Wire Wound (Power Type), Styles RW80 and RW81
MIL-R-93	Resistors, Fixed, Wire-Wound (Accurate), General Specification for
MIL-R-93/10	Resistors, Fixed, Wire-Wound (Accurate), Style RB52
MIL-R-93/11	Resistors, Fixed, Wire-Wound (Accurate), Style RB53
MIL-R-93/12	Resistors, Fixed, Wire-Wound (Accurate), Style RB54
MIL-R-93/13	Resistors, Fixed, Wire-Wound (Accurate), Style RB55
MIL-R-93/14	Resistors, Fixed, Wire-Wound (Accurate), Style RB56
MIL-R-93/19	Resistors, Fixed, Wire-Wound (Accurate), Style RB70
MIL-R-93/20	Resistors, Fixed, Wire-Wound (Accurate), Style RB71
MIL-DTL-6749	Resistor, Variable, Wire Wound, Nonprecision, Aircraft Power
MIL-DTL-7790	Resistor, Thermocouple Lead Spool
MIL-R-10509	Resistors, Fixed, Film (High Stability), General Specification For
MIL-R-10509/1	Resistors, Fixed, Film (High Stability), Style RN60
MIL-R-10509/2	Resistors, Fixed, Film (High Stability), Style RN65
MIL-R-10509/3	Resistors, Fixed, Film (High Stability), Style RN70
MIL-R-10509/7	Resistors, Fixed, Film (High Stability), Style RN55
MIL-R-10509/8	Resistors, Fixed, Film (High Stability), Style RN50
MIL-R-18546/1	Resistors, Fixed, Wire-Wound (Power Type, Chassis Mounted), Styles RE60, RE65, RE70, and RE75
MIL-R-19365	Resistors, Adjustable, Wire-Wound, Power
MIL-PRF-22097/1	Resistors, Variable, Nonwire-Wound (Adjustment Type, Lead-Screw Actuated), Style RJ11
MIL-PRF-22097/3	Resistors, Variable, Nonwire-Wound (Adjustment Type, Lead-Screw Actuated), Style RJ22
MIL-PRF-22097/5	Resistors, Variable, Nonwire-Wound (Adjustment Type, Lead-Screw Actuated), Style RJ26
MIL-PRF-22097/6	Resistors, Variable, Nonwire-Wound (Adjustment Type, Single Turn), Style RJ50

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APPENDIX D

D-I. Inactive DoD specifications. - Continued

Specification number	Title
MIL-PRF-23648/2	Resistor, Thermal (Thermistor), Insulated, Negative Temperature Coefficient, Style RTH08
MIL-PRF-23648/3	Resistor, Thermal (Thermistor), Insulated, Negative Temperature Coefficient, Style RTH10
MIL-PRF-22684/1	Resistors, Fixed, Film, Insulated, Style RL07
MIL-PRF-22684/2	Resistors, Fixed, Film, Insulated, Style RL20
MIL-PRF-22684/3	Resistors, Fixed, Film, Insulated, Style RL32
MIL-PRF-22684/4	Resistors, Fixed, Film, Insulated, Style RL42
MIL-PRF-27208/4	Resistor, Variable, Wire Wound (Adjustment Type, Lead- Screw Actuated) Style RT22
MIL-PRF-39005/7	Resistors, Fixed, Wire-Wound (Accurate), Nonestablished Reliability, and Established Reliability, Style RBR57
MIL-PRF-39005/8	Resistors, Fixed, Wire-Wound (Accurate), Nonestablished Reliability, and Established Reliability, Style RBR74
MIL-PRF-39005/9	Resistors, Fixed, Wire-Wound (Accurate), Nonestablished Reliability, and Established Reliability, Style RBR75
MIL-PRF-39007/5	Resistor, Fixed, Wire-Wound (Power Type), Nonestablished Reliability, Established Reliability, and Space Level, Style RWR71
MIL-PRF-39007/6	Resistor, Fixed, Wire-Wound (Power Type), Nonestablished Reliability, Established Reliability, and Space Level, Style RWR74
MIL-PRF-39015/1	Resistors, Variable, Wire Wound (Lead Screw Actuated), Nonestablished Reliability and Established Reliability, Style RTR12
MIL-PRF-39035/1	Resistors, Variable, Nonwirewound (Adjustment Type, Lead-Screw Actuated), Nonestablished Reliability, and Established Reliability Style RJR12
MIL-PRF-55182/2	Resistors, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level, Style RNR57
MIL-R-55182/11	Resistors, Fixed, Film, Established Reliability Style RNR51
MIL-R-55182/12	Resistors, Fixed, Film, Established Reliability Style RNR56

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APPENDIX D

D.3.2 INACTIVE COMMERCIAL ITEM DESCRIPTIONS (CIDs)

D.3.2.1 Inactive Commercial Item Descriptions (CIDs). Table D-II contains a listing of Inactive Commercial Item Descriptions (CIDs)

D-II. Inactive Commercial Item Descriptions (CIDs). 1/

1/ At this time there are no inactive Commercial Item Descriptions (CIDs).

D.3.3 INACTIVE DSCC DRAWINGS

D.3.3.1 Inactive DSCC Drawings. Table D-III contains a listing of Inactive DSCC Drawings with supersession data.

D-III. Inactive DSCC Drawings.

Drawing Number	Title	Superseding document
03002	Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0505	MIL-PRF-32159/2
03012	Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0302	MIL-PRF-32159/13
03013	Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0603	MIL-PRF-32159/12
03014	Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0402	MIL-PRF-32159/11
03015	Resistor, Chip, Fixed, Film, Zero-Ohm, Style 2010	MIL-PRF-32159/8
03016	Resistor, Chip, Fixed, Film, Zero-Ohm, Style 2512	MIL-PRF-32159/9
03017	Resistor, Thermal, Thermistor, Die Chip, Positive Temperature Coefficient (PTC)	MIL-PRF-32192/1
03018	Resistor, Thermal, Thermistor, Die Chip, Negative Temperature Coefficient (NTC), Style 0404	MIL-PRF-32192/3
87011	Zero-Ohm Chip Resistor, Style 1010	MIL-PRF-32159/10
87033	Resistor Network, 10-Pin, Single-in-Line Package (Sip)	MIL-PRF-83401/24
87063	Resistor, Voltage Sensitive (Varistor)	A-A-55564/3
88032	Resistor, Fixed, Film, Zero-Ohm, Chip, Style RM0502	MIL-PRF-32159/1
88033	Resistor, Fixed, Film, Chip, Style 1010	MIL-PRF-55342/10
90047	Resistor, Fixed, Chip, Zero-Ohm, Style RM2208	MIL-PRF-32159/5
90048	Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0705	MIL-PRF-32159/6
90049	Resistor, Chip, Fixed, Film, Zero-Ohm, Style 1005	MIL-PRF-32159/3
90065	Resistor, Voltage Sensitive (Varistor)	A-A-55562/2
90092	Resistor, Chip, Fixed, Film, Zero-Ohm, Style RM1505	MIL-PRF-32159/4
93073	Resistor, Thermal, Chip, NTC, Style 0805	MIL-PRF-32192/4
93074	Thermistor, (Thermally Sensitive Resistor), NTC, Chip, Style 1206	MIL-PRF-32192/5
94011	Resistor, Chip, Fixed, Film, Zero-Ohm, Style RM1206	MIL-PRF-32159/7
97008	Resistor, Chip, 1/4 Watt (MELF)	05009
98018	Resistor, Chip, Thermal (Thermistor), PTC, Style 0805	MIL-PRF-32192/2

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APPENDIX E

REPLACEMENT PARTS FOR MIL-R-39008 AND MIL-R-11

E.1 SCOPE

E.1.1 Scope. The military specifications MIL-R-39008 (Resistors, Fixed, Composition (Insulated) Established Reliability, General Specification For) and MIL-R-11 (Resistors, Fixed, Composition (Insulated)) have been cancelled. There are no direct replacements available. The following suggested substitutes do not meet all of the requirements of MIL-R-39008 and MIL-R-11.

E.2 APPLICABLE DOCUMENTS. This section is not applicable to this section.

E.3 SUGGESTED REPLACEMENTS

E.3.1 Replacement for general applications. The following specifications are considered acceptable substitutes for general applications of the MIL-R-39008 and MIL-R-11.

Department of Defense Specifications:

MIL-PRF-22684/8	-	Resistor, Fixed, Film, Insulated, Style RL42….TX.
MIL-PRF-39017/1	-	Resistors, Fixed, Film, (Insulated), Nonestablished Reliability, and Established Reliability, Style RLR07.
MIL-PRF-39017/2	-	Resistors, Fixed, Film, (Insulated), Nonestablished Reliability, and Established Reliability, Style RLR20.
MIL-PRF-39017/3	-	Resistors, Fixed, Film, (Insulated), Nonestablished Reliability, and Established Reliability, Style RLR32.
MIL-PRF-39017/5	-	Resistors, Fixed, Film, (Insulated), Nonestablished Reliability, and Established Reliability, Style RLR05.

DSCC drawings:

97004	-	Resistors, Fixed, Film (Insulated), 2 Watt
98020	-	Resistors, Fixed, Film (Insulated), 1/8 Watt
98021	-	Resistors, Fixed, Film (Insulated), 1/2 Watt
98022	-	Resistors, Fixed, Film (Insulated), 1 Watt
99011	-	Resistors, Fixed, Film (Insulated), 0.25 Watt

Designers are CAUTIONED on using these resistors in pulse applications. See E.4.1

E.3.2 Replacement for pulse applications. The following specifications are considered acceptable substitutes for pulse application use of the MIL-R-39008 and MIL-R-11 part.

DSCC drawings:

03003	-	Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/8 Watt
03004	-	Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/4 Watt
03005	-	Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/2 Watt
03006	-	Resistor, Fixed, Carbon Film, High Pulse Voltage, 1 Watt
03007	-	Resistor, Fixed, Carbon Film, High Pulse Voltage, 3/4 Watt
03008	-	Resistor, Fixed, Carbon Film, High Pulse Voltage, 2 Watt
03009	-	Resistor, Fixed, Carbon Film, High Pulse Voltage, 3 Watt
99001	-	Resistor, Fixed, Ceramic Composition (Insulated)

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APPENDIX E

E.4 ADDITIONAL INFORMATION

E.4.1 Fixed film resistors. Designers are CAUTIONED on using these resistors in pulse applications. These resistors only see a one time pulse (Short-time overload) as part of the group B inspection. These resistors are subject to damage and premature failures when they see excessive pulsing.

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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA -CC

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
DLA - CC

(Project 5905-2008-029)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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